Calculation Cover Sheet

1. Document Title:		
Criticality Control Overpack Criticality Analysis of Criticality Control Container Constructed of Carbon Steel		
2. Document Number: CCO-CAL-0008	3. Document Revision: 0	4. Page: 1 of 25

5. Summary Description

A criticality analysis was performed for the criticality control overpack (CCO) transported within either a TRUPACT-II or HalfPACT transportation package as documented in 01937.01.M009-01. In the referenced analysis, the CCC material of construction is stainless steel 304. This analysis evaluates carbon steel construction of the CCC as a fabrication alternative. The calculational model with the highest multiplication factor from the reference is used to determine the impact of this proposed change and to ensure that use of carbon steel for the CCC remains subcritical.

6. Software Usage

Software Name	Version
1. SCALE	6.2.3
2.	
3.	
4.	

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TABLE OF REVISIONS

Revision Number	Pages Affected	Revision Description
0	All	New Issue

1.0 INTRODUCTION

A criticality analysis was performed for the criticality control overpack (CCO) transported within either a TRUPACT-II or HalfPACT transportation package to demonstrate that the package and payload complies with the requirements of 10 CFR 71.55 and 71.59 [1], as documented in 01937.01.M009-01 [2]. A criticality control container (CCC), containing up to 380 fissile gram equivalent (FGE) Pu-239 is transported inside a CCO.

In the above referenced analysis, the CCC material of construction is stainless steel 304. This analysis evaluates carbon steel construction of the CCC as a fabrication alternative. The calculational model with the highest k_{eff} from the reference is used to determine the impact of the proposed change and to ensure that use of carbon steel for the CCC remains subcritical.

2.0 DESCRIPTION OF CRITICALITY DESIGN

2.1.1 Design Features

A CCC is transported inside a CCO. The CCC may be constructed of stainless steel 304 or a proposed alternative of carbon steel and has an inner diameter of 6.065-in, an outer diameter of 6.625-in, and a cavity height of 26.943-in. The CCC confines the fissile material in a critically safe geometry. Also, the CCO maintains a minimum outer diameter of 15.5-in and height of 31.125-in under accident conditions, which provides spacing between the CCCs. The TRUPACT-II may transport 14 CCOs (two layers of 7 CCOs), while the HalfPACT may transport 7 CCOs (single layer). The outer shell of the TRUPACT-II or HalfPACT provides separation between the payloads in adjacent packages.

2.1.2 Summary Table of Criticality Evaluation

Table 2-1 summarizes the maximum results of the criticality calculations for the CCO performed with SCALE 4.4 in 01937.01.M009-01 [2]. The maximum calculated k_{safe} (k_s), which is defined as k-effective (k_{eff}) plus twice the statistical uncertainty (σ), occurs with 380 FGE optimally moderated in each CCO in the TRUPACT-II array model under hypothetical accident conditions (HAC).

The upper subcritical limit (USL) for ensuring that the package is acceptably subcritical, as determined with SCALE version 4.4 in Section 5.0 of the reference was 0.9377. The package is considered to be acceptably subcritical if the computed k_s is less than the USL, or:

$$k_s = k_{eff} + 2\sigma < USL$$

The USL is determined on the basis of a benchmark analysis and incorporates the combined effects of code computational bias, the uncertainty in the bias based on benchmark-model uncertainties, and an administrative margin of 0.05. The results of the benchmark analysis indicate that the USL is adequate to ensure subcriticality of the package. SCALE 4.4 is no longer available for use in this analysis so SCALE 6.2.3 is used herein. The USL is calculated using a similar benchmark analysis with this code version and cross-sections supplied with it as discussed in Section 5.0.

The maximum results of the HAC array of TRUPACT-IIs each containing 14 CCOs with CCCs constructed of stainless steel 304 or carbon steel under SCALE 6.2.3 are given in Table 2-2. The maximum calculated k_s under HAC is higher when the CCC is constructed of carbon steel compared to stainless steel 304, but remains below the USL.

Table 2-1 Summary of CCO Criticality Evaluation with SCALE 4.4 from 01937.01.M009-01

Normal Conditions of Transport (NCT)	
	k_s
Single Unit Maximum	0.8033
Infinite Array Maximum	0.9002
Hypothetical Accident Conditions (HAC)	
	k_s
Single Unit Maximum	0.8209
Infinite Array Maximum	0.9357
USL = 0.9377	

Table 2-2 Summary of CCO HAC Infinite Array Results with SCALE 6.2.3

Hypothetical Accident Conditions (HAC)		
	k_s SS304 CCC	k_s Carbon Steel CCC
Infinite Array Maximum	0.9299	0.9333
USL = 0.9375		

2.1.3 Criticality Safety Index

A criticality safety index (CSI) of 0 is justified, whether the CCC is constructed of stainless steel 304 or carbon steel. The CSI is defined in 10 CFR 71.59 [1] as 50/N, where N = infinite for a CSI = 0. Both the NCT and HAC array cases utilize an infinite number of packages stacked two high.

3.0 COMPUTER CODES AND CROSS-SECTION LIBRARIES

The calculations documented in 01937.01.M009-01 [2] used SCALE 4.4, which is no longer a supported version. Instead, SCALE 6.2.3 with the ENDF/B-VII.1 evaluated nuclear data library in continuous energy (CE_V7.1_ENDF) form [3] installed and verified on DELL 610 16 core server [4] is used herein. The h-poly cross-section has been corrected per the method identified in response to SCALE error bulletin SDL-2021-024 [5]. SCALE 6.2.3 is validated for proper operation on the computer platform by evaluating the same benchmark criticality experiments as

performed in 01937.01.M009-01. Descriptions of the benchmark calculations and validation methodology are provided in Section 5.0.

SCALE 6.2.3 cases are run with 10,000 neutrons per generation for 1,200 generations, skipping the first 200. The 1-sigma uncertainty is approximately 0.00030 for all cases.

4.0 RESULTS

The SCALE input file named d4 that resulted in the highest k_{eff}, as listed in Section 10.2 of 01937.01.M009-01, was rerun in SCALE 6.2.3. The input file was then modified to change the material of the CCC from stainless steel 304 to carbon steel as listed in Appendix 7.1. As shown in Table 4-1, the results of these calculations show that carbon steel results in a higher k_{eff}. However, the results remain below the SCALE 6.2.3 of 0.9375 as determined in Section 5.4. The energy of the average lethargy causing fission (EALF) as calculated by the code is also listed in the table. An electronic copy of the input and output files are available in the electronic document management system.

Table 4-1 Benchmark Experiments Results

Filename	CCC material of construction	EALF (eV)	k_{eff}	σ	k_s
d4	Stainless steel 304	7.27E-02	0.92940	0.00026	0.9299
d4cs	Carbon steel	7.24E-02	0.93276	0.00028	0.9333

5.0 BENCHMARK EVALUATIONS

5.1 Applicability of Benchmark Experiments

The benchmark experiments of water-reflected solutions of plutonium nitrate evaluated in 01937.01.M009-01 [2] are also evaluated herein using the KENO-V.a Monte Carlo criticality code in SCALE version 6.2.3 with continuous energy, ENDF-B/VII.1 cross-section library (CE_V7.1_ENDF). These experiments are presented in Table 5-1. Detailed descriptions of the benchmark experiments are obtained from the OECD Nuclear Energy Agency's *International Handbook of Evaluated Criticality Safety Benchmark Experiments* [6].

Benchmarks are selected for thermal solutions of plutonium and for beryllium in a thermal spectra. For beryllium, the only experimental case that met this requirement is U233-SOL-THERM-015. No cases are available in which beryllium and plutonium are present in a thermal/moderated environment in the same experiment. The benchmark cases are typically fissile solutions that account for the optimally moderated fissile regions of this analysis. Some of the plutonium experiments also contained Plexiglas, which is used to validate the polyethylene, CH_2, which is also a hydrocarbon material.

Table 5-1 Benchmark Experiments Utilized

Series	Title
PU-SOL-THERM-001	Water-Reflected 11.5-Inch Diameter Spheres of Plutonium Nitrate Solutions
PU-SOL-THERM-002	Water-Reflected 12-Inch Diameter Spheres of Plutonium Nitrate Solutions
PU-SOL-THERM-003	Water-Reflected 13-Inch Diameter Spheres of Plutonium Nitrate Solutions
PU-SOL-THERM-004	Water-Reflected 14-Inch Diameter Spheres of Plutonium Nitrate Solutions 0.54% To 3.43% Pu240
PU-SOL-THERM-005	Water-Reflected 14-Inch Diameter Spheres of Plutonium Nitrate Solutions 4.05% And 4.40% Pu240
PU-SOL-THERM-006	Water-Reflected 15-Inch Diameter Spheres of Plutonium Nitrate Solutions
PU-SOL-THERM-007	Water-Reflected 11.5-Inch Diameter Spheres Partly Filled with Plutonium Nitrate Solutions
PU-SOL-THERM-009	Unreflected 48-Inch-Diameter Sphere of Plutonium Nitrate Solution
PU-SOL-THERM-010	Water-Reflected 9-, 10-, 11-, And 12-Inch-Diameter Cylinders of Plutonium Nitrate Solutions
PU-SOL-THERM-011	Bare 16- And 18-Inch-Diameter Spheres of Plutonium Nitrate Solutions
PU-SOL-THERM-014	Interacting Cylinders of 300-mm Diameter Spheres of Plutonium Nitrate Solution (115.1gPu/L) in Air
PU-SOL-THERM-015	Interacting Cylinders of 300-mm Diameter with Plutonium Nitrate Solution (152.5gPu/L) in Air
PU-SOL-THERM-016	Interacting Cylinders of 300-mm and 256-mm Diameters with Plutonium Nitrate Solution (152.5 and 115.1 gPu/L) and Nitric Acid (2n) in Air
PU-SOL-THERM-017	Interacting Cylinders of 256-mm and 300-mm Diameters with Plutonium Nitrate Solution (115.1 gPu/L) in Air
PU-SOL-THERM-020	Water-Reflected and Water-Cadmium Reflected 14-inch Diameter Spheres of Plutonium Nitrate Solutions
PU-SOL-THERM-021	Water-Reflected and Bare 15.2-inch Diameter Spheres of Plutonium Nitrate Solutions
PU-SOL-THERM-024	Slabs of Plutonium Nitrate Solutions Reflected by 1-inch Thick Plexiglas
U233-SOL-THERM-015	Uranyl-Fluoride (^{233}U) Solutions in Spherical Stainless Steel Vessels with Reflectors of Be, CH_2, and Be-CH_2 Composites

5.2 Bias Determination

Table 5-3 shows the experimental benchmark k_{eff} (k_{exp}) and calculated k_{eff} values (k_{calc}) for each Pu solution benchmark experiment. An electronic copy of the input and output files, as listed in Appendix 7.2, are available in the electronic document management system. A histogram of the normalized values $k_{norm}=k_{calc}/k_{exp}$ is shown in Figure 5-1. Three standard tests for normality are applied to the normalized data as summarized in Table 5-2. Since the data fail two of the three normality tests, the distribution is deemed non-normal and the non-parametric method from NUREG/CR-6698 [7] is used to determine the bias and bias uncertainty.

This non-parametric method results in a determination of the degree of confidence that a fraction of the true population of data lies above the smallest observed value. This determination is calculated as follows:

$$\beta = 1 - \sum_{j=0}^{m-1} \frac{n!}{j!\,(n-j)!}(1-q)^j q^{n-j}$$

where

β =level of confidence,

q =the desired population fraction (0.95 for this validation),

n = the number of data in one data sample,

m =the rank order indexing from the smallest sample to the largest ($m = 1$ for the smallest sample). Non-parametric techniques do not require reliance upon distributions, but are rather an analysis of ranks,

j = the ranked sample in the sample population being evaluated.

As stated in NUREG/CR-6698, for a desired population fraction of 95% and a rank order of 1 (the smallest data sample) the equation simplifies to:

$$\beta = 1 - 0.95^n$$

For a non-parametric set of data, the bias and bias uncertainty are determined as follows:

Bias+Bias Uncertainty = 1 – (Smallest k_{eff} value in the data set – S_t – NPM)

where

S_t = total standard deviation corresponding to the smallest k_{norm} value in the data set

$= \sqrt{(\sigma_{exp}{}^2 + \sigma_{calc}{}^2}$

NPM = non-parametric margin, determined from β

With n=196 benchmarks, the level of confidence, β, is essentially unity and the corresponding NPM from NUREG/CR-6998 Table 2.2 is zero.

Hence, the resulting bias and bias uncertainty are determined from the benchmark case with the lowest k_{norm} value, pust011_case_4.18, as

Bias+Bias Uncertainty = 1 – (0.99277 – 0.00521 – 0.0) = 0.01244

Table 5-2 Normality Test Results

Test	Result
Modified Chi-Square Test	Not Normal
Kolmogorov-Smirnov Test	Normal
Lilliefors Test	Not Normal

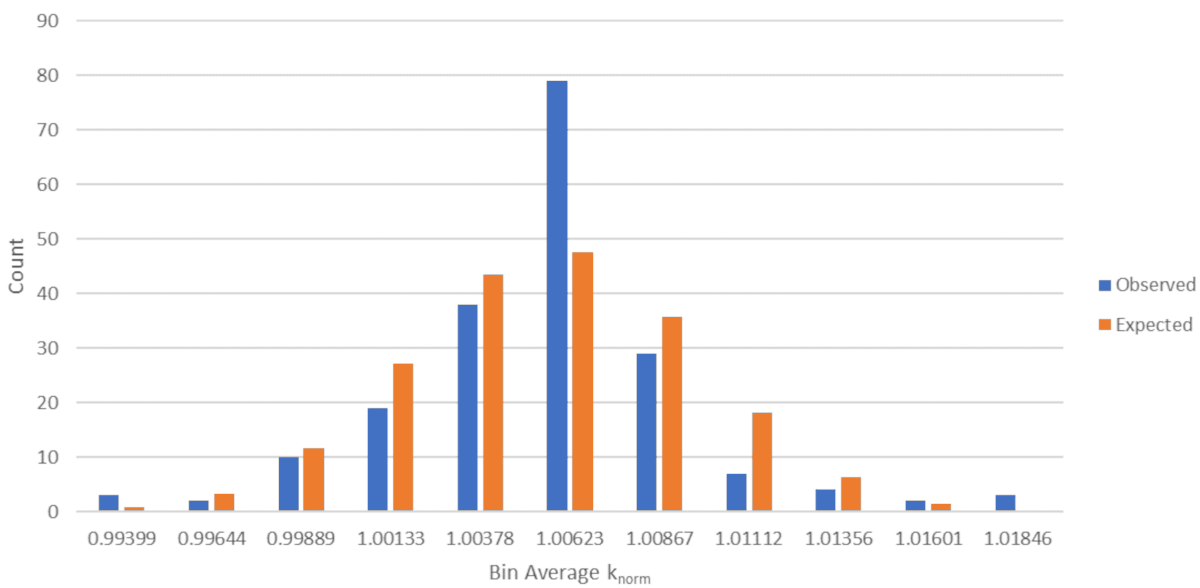

Figure 5-1 Normalized k_{eff} Values for the Pu Benchmark Set

Table 5-3 Benchmark Experiments Results

Case	Experiment			Calculation			Normalized	
	k_{exp}	σ_{exp}	H/Pu	k_{calc}	σ_{calc}	EALF [eV]	k_{norm}	σ_{total}
pust001_case_1	1.00000	0.00500	371	1.00550	0.00031	8.72E-02	1.00550	0.00501
pust001_case_2	1.00000	0.00500	272	1.00753	0.00030	1.10E-01	1.00753	0.00501
pust001_case_3	1.00000	0.00500	216	1.01031	0.00029	1.33E-01	1.01031	0.00501
pust001_case_4	1.00000	0.00500	190	1.00477	0.00030	1.49E-01	1.00477	0.00501
pust001_case_5	1.00000	0.00500	180	1.00878	0.00031	1.57E-01	1.00878	0.00501
pust001_case_6	1.00000	0.00500	91	1.01021	0.00030	3.41E-01	1.01021	0.00501
pust002_case_1	1.00000	0.00470	524	1.00388	0.00027	7.10E-02	1.00388	0.00471
pust002_case_2	1.00000	0.00470	505	1.00455	0.00027	7.26E-02	1.00455	0.00471
pust002_case_3	1.00000	0.00470	451	1.00275	0.00028	7.75E-02	1.00275	0.00471
pust002_case_4	1.00000	0.00470	421	1.00591	0.00027	8.09E-02	1.00591	0.00471
pust002_case_5	1.00000	0.00470	393	1.00921	0.00029	8.45E-02	1.00921	0.00471
pust002_case_6	1.00000	0.00470	344	1.00466	0.00032	9.22E-02	1.00466	0.00471
pust002_case_7	1.00000	0.00470	309	1.00734	0.00029	9.94E-02	1.00734	0.00471
pust003_case_1	1.00000	0.00470	788	1.00224	0.00030	5.84E-02	1.00224	0.00471
pust003_case_2	1.00000	0.00470	756	1.00175	0.00027	5.95E-02	1.00175	0.00471
pust003_case_3	1.00000	0.00470	699	1.00444	0.00030	6.19E-02	1.00444	0.00471
pust003_case_4	1.00000	0.00470	682	1.00388	0.00026	6.27E-02	1.00388	0.00471
pust003_case_5	1.00000	0.00470	627	1.00523	0.00026	6.53E-02	1.00523	0.00471
pust003_case_6	1.00000	0.00470	563	1.00532	0.00030	6.91E-02	1.00532	0.00471
pust003_case_7	1.00000	0.00470	738	1.00628	0.00027	5.92E-02	1.00628	0.00471
pust003_case_8	1.00000	0.00470	714	1.00521	0.00029	6.02E-02	1.00521	0.00471
pust004_case_1	1.00000	0.00470	987	1.00312	0.00026	5.35E-02	1.00312	0.00471
pust004_case_10	1.00000	0.00470	689	1.00138	0.00028	6.31E-02	1.00138	0.00471
pust004_case_11	1.00000	0.00470	592	1.00053	0.00027	6.83E-02	1.00053	0.00471
pust004_case_12	1.00000	0.00470	893	1.00267	0.00031	5.59E-02	1.00267	0.00471
pust004_case_13	1.00000	0.00470	903	1.00007	0.00027	5.57E-02	1.00007	0.00471
pust004_case_2	1.00000	0.00470	977	0.99819	0.00028	5.38E-02	0.99819	0.00471
pust004_case_3	1.00000	0.00470	935	1.00034	0.00028	5.48E-02	1.00034	0.00471
pust004_case_4	1.00000	0.00470	889	0.99809	0.00025	5.61E-02	0.99809	0.00471
pust004_case_5	1.00000	0.00470	942	0.99957	0.00027	5.46E-02	0.99957	0.00471
pust004_case_6	1.00000	0.00470	927	1.00139	0.00026	5.51E-02	1.00139	0.00471
pust004_case_7	1.00000	0.00470	892	1.00522	0.00032	5.59E-02	1.00522	0.00471
pust004_case_8	1.00000	0.00470	869	1.00060	0.00029	5.67E-02	1.00060	0.00471
pust004_case_9	1.00000	0.00470	805	1.00013	0.00028	5.86E-02	1.00013	0.00471
pust005_case_1	1.00000	0.00470	903	1.00152	0.00027	5.57E-02	1.00152	0.00471
pust005_case_2	1.00000	0.00470	868	1.00215	0.00027	5.67E-02	1.00215	0.00471
pust005_case_3	1.00000	0.00470	834	1.00373	0.00026	5.77E-02	1.00373	0.00471
pust005_case_4	1.00000	0.00470	765	1.00437	0.00027	6.01E-02	1.00437	0.00471
pust005_case_5	1.00000	0.00470	694	1.00595	0.00028	6.29E-02	1.00595	0.00471
pust005_case_6	1.00000	0.00470	633	1.00538	0.00027	6.59E-02	1.00538	0.00471
pust005_case_7	1.00000	0.00470	581	1.00361	0.00026	6.90E-02	1.00361	0.00471
pust005_case_8	1.00000	0.00470	869	0.99896	0.00028	5.67E-02	0.99896	0.00471
pust005_case_9	1.00000	0.00470	825	1.00175	0.00025	5.80E-02	1.00175	0.00471
pust006_case_1	1.00000	0.00350	1061	1.00025	0.00030	5.26E-02	1.00025	0.00351
pust006_case_2	1.00000	0.00350	1018	1.00119	0.00027	5.35E-02	1.00119	0.00351
pust006_case_3	1.00000	0.00350	940	1.00109	0.00029	5.53E-02	1.00109	0.00351

Case	Experiment			Calculation			Normalized	
	k_{exp}	σ_{exp}	H/Pu	k_{calc}	σ_{calc}	EALF [eV]	k_{norm}	σ_{total}
pust007_case_10	1.00000	0.00470	285	1.00035	0.00032	1.05E-01	1.00035	0.00471
pust007_case_2	1.00000	0.00470	110	1.00927	0.00030	2.70E-01	1.00927	0.00471
pust007_case_3	1.00000	0.00470	114	1.00355	0.00033	2.56E-01	1.00355	0.00471
pust007_case_5	1.00000	0.00470	268	1.00882	0.00031	1.11E-01	1.00882	0.00471
pust007_case_6	1.00000	0.00470	262	1.00322	0.00031	1.13E-01	1.00322	0.00471
pust007_case_7	1.00000	0.00470	266	1.00549	0.00029	1.11E-01	1.00549	0.00471
pust007_case_8	1.00000	0.00470	258	0.99905	0.00028	1.14E-01	0.99905	0.00471
pust007_case_9	1.00000	0.00470	260	0.99659	0.00028	1.13E-01	0.99659	0.00471
pust009_case_1	1.00000	0.00330	2648	1.01470	0.00026	4.12E-02	1.01470	0.00331
pust009_case_2	1.00000	0.00330	2779	1.01968	0.00022	4.07E-02	1.01968	0.00331
pust009_case_3	1.00000	0.00330	2803	1.01827	0.00023	4.06E-02	1.01827	0.00331
pust010_case_1.11	1.00000	0.00480	485	1.01221	0.00030	7.51E-02	1.01221	0.00481
pust010_case_1.12	1.00000	0.00480	543	1.00904	0.00029	6.97E-02	1.00904	0.00481
pust010_case_1.9	1.00000	0.00480	267	1.01792	0.00028	1.09E-01	1.01792	0.00481
pust010_case_2.11	1.00000	0.00480	558	1.01006	0.00030	6.97E-02	1.01006	0.00481
pust010_case_2.12	1.00000	0.00480	618	1.00895	0.00028	6.54E-02	1.00895	0.00481
pust010_case_2.9	1.00000	0.00480	357	1.01410	0.00033	8.88E-02	1.01410	0.00481
pust010_case_3.11	1.00000	0.00480	558	1.00927	0.00029	6.87E-02	1.00927	0.00481
pust010_case_3.12	1.00000	0.00480	728	1.01560	0.00034	6.08E-02	1.01560	0.00481
pust010_case_3.9	1.00000	0.00480	484	1.00776	0.00033	7.38E-02	1.00776	0.00481
pust010_case_4.11	1.00000	0.00480	606	1.00205	0.00028	6.64E-02	1.00205	0.00481
pust010_case_4.12	1.00000	0.00480	850	1.00993	0.00026	5.71E-02	1.00993	0.00481
pust010_case_5.11	1.00000	0.00480	665	1.00334	0.00028	6.40E-02	1.00334	0.00481
pust010_case_6.11	1.00000	0.00480	414	1.01437	0.00026	8.10E-02	1.01437	0.00481
pust010_case_7.11	1.00000	0.00480	535	1.00227	0.00029	7.03E-02	1.00227	0.00481
pust011_case_1.16	1.00000	0.00520	765	1.00917	0.00030	6.34E-02	1.00917	0.00521
pust011_case_1.18	1.00000	0.00520	1208	0.99387	0.00027	5.22E-02	0.99387	0.00521
pust011_case_2.16	1.00000	0.00520	736	1.01353	0.00033	6.47E-02	1.01353	0.00521
pust011_case_2.18	1.00000	0.00520	1151	0.99943	0.00030	5.32E-02	0.99943	0.00521
pust011_case_3.16	1.00000	0.00520	691	1.01619	0.00033	6.69E-02	1.01619	0.00521
pust011_case_3.18	1.00000	0.00520	1158	0.99582	0.00028	5.31E-02	0.99582	0.00521
pust011_case_4.16	1.00000	0.00520	682	1.00835	0.00029	6.76E-02	1.00835	0.00521
pust011_case_4.18	1.00000	0.00520	1100	0.99277	0.00027	5.44E-02	0.99277	0.00521
pust011_case_5.16	1.00000	0.00520	575	1.00557	0.00028	7.52E-02	1.00557	0.00521
pust011_case_5.18	1.00000	0.00520	1039	1.00301	0.00029	5.58E-02	1.00301	0.00521
pust011_case_6.18	1.00000	0.00520	908	0.99942	0.00026	5.94E-02	0.99942	0.00521
pust011_case_7.18	1.00000	0.00520	1103	0.99883	0.00029	5.41E-02	0.99883	0.00521
pust014_case_1	0.99800	0.00320	220	1.00489	0.00029	1.65E-01	1.00690	0.00321
pust014_case_10	0.99800	0.00320	220	1.00370	0.00031	1.64E-01	1.00571	0.00321
pust014_case_11	0.99800	0.00320	220	1.00336	0.00030	1.64E-01	1.00537	0.00321
pust014_case_12	0.99800	0.00320	220	1.00401	0.00032	1.64E-01	1.00602	0.00322
pust014_case_13	0.99800	0.00430	220	1.00524	0.00033	1.66E-01	1.00725	0.00431
pust014_case_14	0.99800	0.00430	220	1.00343	0.00034	1.65E-01	1.00544	0.00431
pust014_case_15	0.99800	0.00430	220	1.00363	0.00031	1.65E-01	1.00564	0.00431
pust014_case_16	0.99800	0.00430	220	1.00371	0.00030	1.64E-01	1.00572	0.00431
pust014_case_17	0.99800	0.00430	220	1.00294	0.00031	1.64E-01	1.00495	0.00431
pust014_case_18	0.99800	0.00430	220	1.00634	0.00030	1.66E-01	1.00836	0.00431

Case	Experiment			Calculation			Normalized	
	k_{exp}	σ_{exp}	H/Pu	k_{calc}	σ_{calc}	EALF [eV]	k_{norm}	σ_{total}
pust014_case_19	0.99800	0.00430	220	1.00377	0.00034	1.65E-01	1.00578	0.00431
pust014_case_20	0.99800	0.00430	220	1.00390	0.00029	1.65E-01	1.00591	0.00431
pust014_case_21	0.99800	0.00430	220	1.00432	0.00032	1.64E-01	1.00633	0.00431
pust014_case_22	0.99800	0.00430	220	1.00394	0.00029	1.64E-01	1.00595	0.00431
pust014_case_23	0.99800	0.00430	220	1.00294	0.00032	1.64E-01	1.00495	0.00431
pust014_case_24	0.99800	0.00430	220	1.00650	0.00032	1.66E-01	1.00852	0.00431
pust014_case_25	0.99800	0.00430	220	1.00241	0.00032	1.66E-01	1.00442	0.00431
pust014_case_26	0.99800	0.00430	220	1.00313	0.00033	1.65E-01	1.00514	0.00431
pust014_case_27	0.99800	0.00430	220	1.00290	0.00029	1.64E-01	1.00491	0.00431
pust014_case_28	0.99800	0.00430	220	1.00329	0.00031	1.64E-01	1.00530	0.00431
pust014_case_29	0.99800	0.00430	220	1.00202	0.00032	1.64E-01	1.00403	0.00431
pust014_case_3	0.99800	0.00320	220	1.00346	0.00035	1.65E-01	1.00547	0.00322
pust014_case_30	0.99800	0.00430	220	1.00395	0.00029	1.66E-01	1.00596	0.00431
pust014_case_31	0.99800	0.00430	220	1.00209	0.00035	1.66E-01	1.00410	0.00431
pust014_case_33	0.99800	0.00430	220	1.00322	0.00030	1.65E-01	1.00523	0.00431
pust014_case_34	0.99800	0.00430	220	1.00315	0.00030	1.65E-01	1.00516	0.00431
pust014_case_4	0.99800	0.00320	220	1.00406	0.00033	1.65E-01	1.00607	0.00322
pust014_case_5	0.99800	0.00320	220	1.00436	0.00034	1.64E-01	1.00637	0.00322
pust014_case_6	0.99800	0.00320	220	1.00401	0.00031	1.64E-01	1.00602	0.00321
pust014_case_7	0.99800	0.00320	220	1.00426	0.00030	1.66E-01	1.00627	0.00321
pust014_case_8	0.99800	0.00320	220	1.00281	0.00041	1.65E-01	1.00482	0.00323
pust014_case_9	0.99800	0.00320	220	1.00315	0.00035	1.65E-01	1.00516	0.00322
pust015_case_1	0.99800	0.00380	163	1.00493	0.00032	2.33E-01	1.00694	0.00381
pust015_case_10	0.99710	0.00470	163	1.00339	0.00035	2.32E-01	1.00631	0.00471
pust015_case_11	0.99710	0.00470	163	1.00257	0.00031	2.34E-01	1.00549	0.00471
pust015_case_12	0.99710	0.00470	163	1.00386	0.00034	2.33E-01	1.00678	0.00471
pust015_case_13	0.99710	0.00470	163	1.00446	0.00032	2.32E-01	1.00738	0.00471
pust015_case_14	0.99710	0.00470	163	1.00525	0.00031	2.32E-01	1.00817	0.00471
pust015_case_15	0.99710	0.00470	163	1.00781	0.00032	2.35E-01	1.01074	0.00471
pust015_case_16	0.99710	0.00470	163	1.00438	0.00037	2.34E-01	1.00730	0.00471
pust015_case_17	0.99710	0.00470	163	1.00457	0.00031	2.33E-01	1.00749	0.00471
pust015_case_2	0.99800	0.00380	163	1.00564	0.00032	2.33E-01	1.00766	0.00381
pust015_case_3	0.99800	0.00380	163	1.00464	0.00033	2.32E-01	1.00665	0.00381
pust015_case_4	0.99800	0.00380	163	1.00588	0.00030	2.32E-01	1.00790	0.00381
pust015_case_5	0.99800	0.00380	163	1.00538	0.00030	2.32E-01	1.00739	0.00381
pust015_case_6	0.99800	0.00380	163	1.00657	0.00032	2.32E-01	1.00859	0.00381
pust015_case_7	0.99710	0.00470	163	1.00720	0.00030	2.34E-01	1.01013	0.00471
pust015_case_8	0.99710	0.00470	163	1.00518	0.00034	2.34E-01	1.00810	0.00471
pust015_case_9	0.99710	0.00470	163	1.00496	0.00033	2.33E-01	1.00788	0.00471
pust016_case_1	0.99800	0.00430	163	1.00430	0.00031	2.34E-01	1.00631	0.00431
pust016_case_10	0.99630	0.00330	220	1.00352	0.00030	1.64E-01	1.00725	0.00331
pust016_case_11	0.99630	0.00330	220	1.00416	0.00033	1.64E-01	1.00789	0.00332
pust016_case_2	0.99800	0.00430	163	1.00429	0.00031	2.33E-01	1.00630	0.00431
pust016_case_3	0.99800	0.00430	163	1.00570	0.00032	2.32E-01	1.00772	0.00431
pust016_case_4	0.99800	0.00430	163	1.00490	0.00034	2.32E-01	1.00691	0.00431
pust016_case_5	0.99690	0.00380	220	1.00344	0.00034	1.65E-01	1.00656	0.00382
pust016_case_6	0.99690	0.00380	220	1.00330	0.00031	1.65E-01	1.00642	0.00381

Case	Experiment			Calculation			Normalized	
	k_{exp}	σ_{exp}	H/Pu	k_{calc}	σ_{calc}	EALF [eV]	k_{norm}	σ_{total}
pust016_case_7	0.99690	0.00380	220	1.00377	0.00035	1.64E-01	1.00689	0.00382
pust016_case_8	0.99690	0.00380	220	1.00465	0.00031	1.65E-01	1.00777	0.00381
pust016_case_9	0.99630	0.00330	220	1.00343	0.00032	1.63E-01	1.00716	0.00332
pust017_case_1	0.99690	0.00380	220	1.00295	0.00034	1.64E-01	1.00607	0.00382
pust017_case_10	0.99690	0.00380	220	1.00271	0.00034	1.65E-01	1.00583	0.00382
pust017_case_11	0.99690	0.00380	220	1.00315	0.00031	1.64E-01	1.00627	0.00381
pust017_case_12	0.99690	0.00380	220	1.00400	0.00033	1.65E-01	1.00712	0.00381
pust017_case_13	0.99690	0.00380	220	1.00299	0.00033	1.65E-01	1.00611	0.00381
pust017_case_14	0.99690	0.00380	220	1.00416	0.00031	1.65E-01	1.00728	0.00381
pust017_case_15	0.99690	0.00380	220	1.00394	0.00031	1.64E-01	1.00706	0.00381
pust017_case_16	0.99690	0.00380	220	1.00410	0.00034	1.64E-01	1.00722	0.00382
pust017_case_17	0.99690	0.00380	220	1.00333	0.00031	1.65E-01	1.00645	0.00381
pust017_case_18	0.99690	0.00380	220	1.00340	0.00035	1.65E-01	1.00652	0.00382
pust017_case_2	0.99690	0.00380	220	1.00275	0.00032	1.65E-01	1.00587	0.00381
pust017_case_3	0.99690	0.00380	220	1.00301	0.00032	1.65E-01	1.00613	0.00381
pust017_case_4	0.99690	0.00380	220	1.00291	0.00033	1.65E-01	1.00603	0.00381
pust017_case_5	0.99690	0.00380	220	1.00286	0.00035	1.65E-01	1.00598	0.00382
pust017_case_6	0.99690	0.00380	220	1.00299	0.00031	1.65E-01	1.00611	0.00381
pust017_case_7	0.99690	0.00380	220	1.00257	0.00034	1.65E-01	1.00569	0.00382
pust017_case_8	0.99690	0.00380	220	1.00331	0.00030	1.64E-01	1.00643	0.00381
pust017_case_9	0.99690	0.00380	220	1.00306	0.00033	1.65E-01	1.00618	0.00381
pust020_case_1	1.00000	0.00590	635	1.00356	0.00029	6.55E-02	1.00356	0.00591
pust020_case_2	1.00000	0.00590	654	1.00554	0.00033	6.47E-02	1.00554	0.00591
pust020_case_3	1.00000	0.00590	786	1.00062	0.00027	5.89E-02	1.00062	0.00591
pust020_case_5	1.00000	0.00590	487	1.00400	0.00030	7.59E-02	1.00400	0.00591
pust020_case_6	1.00000	0.00590	476	1.00464	0.00028	7.88E-02	1.00464	0.00591
pust020_case_7	1.00000	0.00590	762	0.99782	0.00028	6.04E-02	0.99782	0.00591
pust020_case_8	1.00000	0.00590	361	1.00278	0.00032	1.05E-01	1.00278	0.00591
pust020_case_9	1.00000	0.00590	568	0.99505	0.00029	7.56E-02	0.99505	0.00591
pust021_case_10	1.00000	0.00250	1159	1.00566	0.00029	5.01E-02	1.00566	0.00252
pust021_case_7	1.00000	0.00320	702	1.00576	0.00028	6.59E-02	1.00576	0.00321
pust021_case_8	1.00000	0.00650	131	1.00300	0.00036	3.07E-01	1.00300	0.00651
pust021_case_9	1.00000	0.00320	667	1.00661	0.00028	6.78E-02	1.00661	0.00321
pust024_case_1	1	0.0062	115	1.00330	0.00031	4.61E-01	1.00330	0.00621
pust024_case_10	1	0.0053	190	1.00913	0.00031	2.22E-01	1.00913	0.00531
pust024_case_11	1	0.0053	190	1.00866	0.00030	2.21E-01	1.00866	0.00531
pust024_case_12	1	0.0053	190	1.00825	0.00030	2.20E-01	1.00825	0.00531
pust024_case_13	1	0.0053	190	1.00741	0.00029	2.20E-01	1.00741	0.00531
pust024_case_14	1	0.0053	161	1.00335	0.00030	2.95E-01	1.00335	0.00531
pust024_case_15	1	0.0053	161	1.00334	0.00032	2.93E-01	1.00334	0.00531
pust024_case_16	1	0.0053	161	1.00438	0.00032	2.91E-01	1.00438	0.00531
pust024_case_17	1	0.0053	161	1.00422	0.00035	2.88E-01	1.00422	0.00531
pust024_case_18	1	0.0051	484	1.00514	0.00030	8.92E-02	1.00514	0.00511
pust024_case_19	1	0.0051	484	1.00657	0.00029	8.90E-02	1.00657	0.00511
pust024_case_2	1	0.0062	115	1.00292	0.00033	4.54E-01	1.00292	0.00621
pust024_case_20	1	0.0051	484	1.00683	0.00030	8.87E-02	1.00683	0.00511
pust024_case_21	1	0.0051	484	1.00618	0.00030	8.85E-02	1.00618	0.00511

Case	Experiment			Calculation			Normalized	
	k_{exp}	σ_{exp}	H/Pu	k_{calc}	σ_{calc}	EALF [eV]	k_{norm}	σ_{total}
pust024_case_22	1	0.0051	484	1.00545	0.00029	8.81E-02	1.00545	0.00511
pust024_case_23	1	0.0051	484	1.00414	0.00029	8.81E-02	1.00414	0.00511
pust024_case_3	1	0.0062	115	1.00350	0.00030	4.53E-01	1.00350	0.00621
pust024_case_4	1	0.0062	115	1.00308	0.00031	4.49E-01	1.00308	0.00621
pust024_case_5	1	0.0062	115	1.00248	0.00031	4.48E-01	1.00248	0.00621
pust024_case_6	1	0.0077	59	1.00266	0.00032	1.64E+00	1.00266	0.00771
pust024_case_7	1	0.0053	190	1.00969	0.00031	2.26E-01	1.00969	0.00531
pust024_case_8	1	0.0053	190	1.00970	0.00034	2.24E-01	1.00970	0.00531
pust024_case_9	1	0.0053	190	1.00884	0.00030	2.23E-01	1.00884	0.00531

5.3 AoA Margin

5.3.1 Beryllium Reflector

Calculated k_{eff} values for cases from the 233-SOL-THERM-015 benchmark experiment are shown in Table 5-4. The benchmark model k_{eff}, k_{exp}, for each case is 1.00000. In Figure 5-2, the results of the beryllium-reflected cases are plotted against EALF and compared to the average calculated k_{eff} of the four non-Be reflected cases in the benchmark set. Although there is a definite trend in the bias at higher EALF values, at the EALF value of interest in this application (approximately 0.0725 eV), the figure shows that the code is over-predicting reactivity relative to the non-Be reflected cases.

Considering this over-prediction in the EALF range of interest, and the low concentration of beryllium in the application considered here (1% by volume), no AoA margin is necessary.

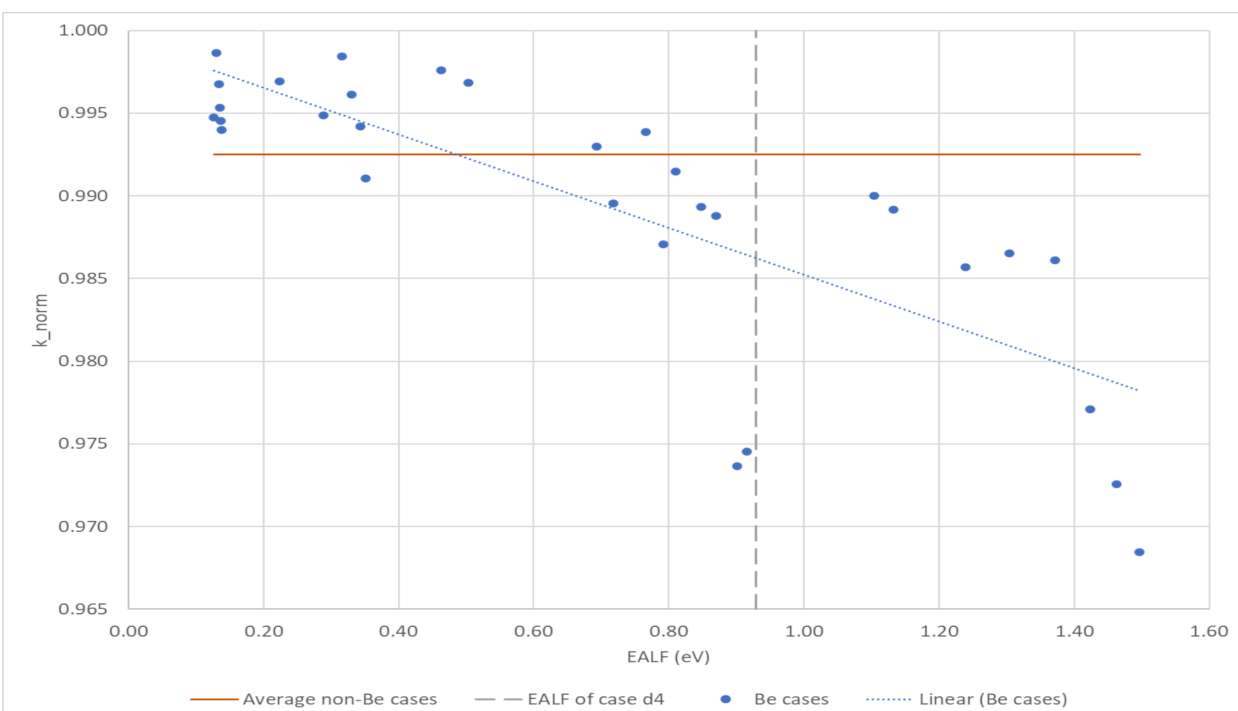

Figure 5-2 k_{norm} Values for the Be-reflected Benchmarks Compared to non-Be Average

Table 5-4 Experiments with and without Beryllium Reflector

Case	Experiment		Calculation			Normalized		Reflector
	k_{exp}	σ_{exp}	k_{calc}	σ_{calc}	EALF [eV]	k_{norm}	σ_{total}	
233st015_case_1	1.00000	0.00750	0.98999	0.00034	1.10E+00	0.98999	0.00751	Be
233st015_case_2	1.00000	0.00700	0.98567	0.00042	1.24E+00	0.98567	0.00701	Be
233st015_case_3	1.00000	0.00680	0.98654	0.00037	1.30E+00	0.98654	0.00681	Be
233st015_case_4	1.00000	0.00410	0.98953	0.00034	7.17E-01	0.98953	0.00411	Be+Poly
233st015_case_5	1.00000	0.00550	0.98610	0.00038	1.37E+00	0.98610	0.00551	Be
233st015_case_6	1.00000	0.00990	0.97711	0.00038	1.42E+00	0.97711	0.00991	Be
233st015_case_7	1.00000	0.00700	0.98708	0.00034	7.92E-01	0.98708	0.00701	Poly
233st015_case_8	1.00000	0.00670	0.97258	0.00037	1.46E+00	0.97258	0.00671	Be
233st015_case_9	1.00000	0.00500	0.96846	0.00036	1.50E+00	0.96846	0.00501	Be
233st015_case_10	1.00000	0.00510	0.98915	0.00036	1.13E+00	0.98915	0.00511	Poly
233st015_case_11	1.00000	0.00750	0.99300	0.00035	6.93E-01	0.99300	0.00751	Be
233st015_case_12	1.00000	0.00690	0.99386	0.00039	7.65E-01	0.99386	0.00691	Be
233st015_case_13	1.00000	0.00690	0.99146	0.00035	8.10E-01	0.99146	0.00691	Be
233st015_case_14	1.00000	0.00360	0.99758	0.00035	4.63E-01	0.99758	0.00362	Be+Poly
233st015_case_15	1.00000	0.00600	0.98932	0.00035	8.47E-01	0.98932	0.00601	Be
233st015_case_16	1.00000	0.00430	0.98880	0.00036	8.69E-01	0.98880	0.00432	Be
233st015_case_17	1.00000	0.00290	0.99685	0.00036	5.03E-01	0.99685	0.00292	Poly
233st015_case_18	1.00000	0.00560	0.97364	0.00036	9.01E-01	0.97364	0.00561	Be
233st015_case_19	1.00000	0.00520	0.97453	0.00037	9.15E-01	0.97453	0.00521	Be
233st015_case_20	1.00000	0.00790	0.99488	0.00036	2.89E-01	0.99488	0.00791	Be
233st015_case_21	1.00000	0.00700	0.99845	0.00037	3.16E-01	0.99845	0.00701	Be
233st015_case_22	1.00000	0.00620	0.99613	0.00038	3.30E-01	0.99613	0.00621	Be
233st015_case_23	1.00000	0.00550	0.99421	0.00038	3.43E-01	0.99421	0.00551	Be
233st015_case_24	1.00000	0.00510	0.99104	0.00034	3.51E-01	0.99104	0.00511	Be
233st015_case_25	1.00000	0.00230	0.99692	0.00037	2.23E-01	0.99692	0.00233	Poly
233st015_case_26	1.00000	0.00660	0.99475	0.00034	1.26E-01	0.99475	0.00661	Be
233st015_case_27	1.00000	0.00630	0.99866	0.00034	1.31E-01	0.99866	0.00631	Be
233st015_case_28	1.00000	0.00580	0.99675	0.00034	1.33E-01	0.99675	0.00581	Be
233st015_case_29	1.00000	0.00510	0.99533	0.00034	1.35E-01	0.99533	0.00511	Be
233st015_case_30	1.00000	0.00480	0.99454	0.00034	1.37E-01	0.99454	0.00481	Be
233st015_case_31	1.00000	0.00550	0.99399	0.00037	1.38E-01	0.99399	0.00551	Be

5.4 USL Determination

The USL is calculated to be 0.9375 (conservatively rounded down from calculated value of 0.93756) as shown in Table 5-5 and determined by the following formula.

USL = 1 – (Bias+Bias Uncertainty) – Administrative Margin – AoA Margin

Table 5-5 Calculation of USL

Benchmark Set	Number of Cases	Bias and Bias Uncertainty	Administrative Margin	AoA Margin	USL
Pu	196	0.01244	0.05	0.00	0.9375

6.0 REFERENCES

1. Title 10, Code of Federal Regulations, Part 71 (10 CFR 71), *Packaging and Transportation of Radioactive Material*.
2. R.J. Migliore, *CCO Criticality Analysis for TRUPACT-II and HalfPACT*, 01937.01.M009-01, Rev. 0, February 2012, AREVA Federal Services LLC.
3. B.T. Rearden and M.A. Jessee, ed., *SCALE Code System*, ORNL/TM-2005/39 Version 6.2.3, March 2018.
4. C.S. Henkel, SCALE Test Verification Forms from STP-PLN-0005, executed on Jan 15, 2020.
5. W. Wieselquist, *SCALE USER NOTICE: SCALE h1-poly incoherent elastic scattering can cause k-eff underprediction approaching 1% in criticality problems with polyethylene as the primary moderator*, Feb 26, 2021, corrected file available at https://www.ornl.gov/content/scale-v624.
6 Organization for Economic Cooperation and Development, *International Handbook of Evaluated Criticality Safety Benchmark Experiments*, NEA/NCS/DOC (95)03, May 2020.
7. J.C. Dean and R.W. Tayloe, *Guide for Validation of Nuclear Criticality Safety Calculational Methodology*, NUREG/CR-6698, January 2001.

7.0 APPENDIX

7.1 CCO Array Input File with Carbon Steel CCC

```
' Scale 6.2.3 with h-poly patch
=shell
copy %SCALEPATH%\..\updates\sdl-2021-024-v1.0.3\fixed_poly\* .
copy ce_v7.1_endf_poly_fix.xml ce_fixed.xml
end

=CSAS5
HAC H/X= 650 Pu(g)= 380 wf 1 D(in)= 15.5 wih CS CCC
ce_fixed
read composition
'H/X= 650
pu-239 1 0 1.0627E-04 293 end
h-poly 1 0 1.9707E-02 293 end
h 1 0 4.9367E-02 293 end
bebound 1 0 1.2322E-03 293 end
c 1 0 9.8535E-03 293 end
o 1 0 2.4684E-02 293 end
'Pure reflector
h-poly 2 0 1.9749E-02 293 end
h 2 0 4.9472E-02 293 end
bebound 2 0 1.2348E-03 293 end
c 2 0 9.8745E-03 293 end
o 2 0 2.4736E-02 293 end
'stainless steel
ss304 3 1.000 END
'carbon steel for CCC
carbonsteel 6 1.000 END
'water
h2o 4 1.0 end
'foam
polyethylene 5 0.1435 end
END COMPosition
'
READ PARAMETERS
gen=1200 npg=10000 nsk=200 htm=no tba=3
END PARAMETERS
'
READ GEOM
'
unit 1
zcylinder 1 1 7.70255 48.3293 0
zcylinder 0 1 7.70255 68.4352 0
zcylinder 6 1 8.4138 70.9752 -2.54
zcylinder 0 1 19.685 73.7464 -5.3111
'
unit 2
```

```
zcylinder 0 1 7.70255 20.1059 0
zcylinder 1 1 7.70255 68.4352 0
zcylinder 6 1 8.4138 70.9752 -2.54
zcylinder 0 1 19.685 73.7464 -5.3111
'
global unit 99
zcylinder 0 1 92.0000 95.0000 -95.0000
hole 1 -19.6851 -69.5744 -10.6294
hole 1 19.6851 -69.5744 -10.6294
hole 1 -39.3702 -35.4788 -10.6294
hole 1 0 -35.4788 -10.6294
hole 1 39.3702 -35.4788 -10.6294
hole 1 -19.6851 -1.3832 -10.6294
hole 1 19.6851 -1.3832 -10.6294
' 0
hole 2 -19.6851 -69.5744 -89.688
hole 2 19.6851 -69.5744 -89.688
hole 2 -39.3702 -35.4788 -89.688
hole 2 0 -35.4788 -89.688
hole 2 39.3702 -35.4788 -89.688
hole 2 -19.6851 -1.3832 -89.688
hole 2 19.6851 -1.3832 -89.688
'
zcylinder 3 1 93.1100 96.2700 -96.2700
zcylinder 0 1 105.8100 108.9700 -108.9700
zcylinder 3 1 106.4050 109.6050 -109.6050
cuboid 0 1 2p 106.4050 2p 106.4050 109.6050 -109.6050
cuboid 4 1 2p 106.4050 2p 106.4050 139.6050 -109.6050
END GEOM
READ BNDS XYF=mir -ZB=mir END BNDS
END DATA
end
```

7.2 List of Files

Below is a list of the SCALE input, message, and output files included in this calculation. The files are available electronically in EDMS.

CCO-CAL-0008 Rev 0 Attachment 1 Directories: 1, Files: 687, Size: 153,533,448 B			
Name	Size	Last write date	CRC32
\Run\ (687)	153,533,448	08-15-2022 07:09	
233st015_case_1.inp	543	07-19-2022 13:56	c6bba109
233st015_case_1.msg	1,490	07-26-2022 03:04	bfbbbcf6
233st015_case_1.out	658,445	07-26-2022 03:47	cecc1342
233st015_case_2.inp	547	07-19-2022 13:56	2cab93e9
233st015_case_2.msg	1,490	07-26-2022 03:04	be0fb927
233st015_case_2.out	658,085	07-26-2022 03:42	57e20f7f
233st015_case_3.inp	543	07-19-2022 13:56	d93ffc4a
233st015_case_3.msg	1,490	07-26-2022 03:04	176040b1
233st015_case_3.out	658,150	07-26-2022 03:40	d64ceb96
233st015_case_4.inp	773	07-19-2022 13:58	ac48e5df
233st015_case_4.msg	1,653	07-26-2022 03:04	356579b7
233st015_case_4.out	662,112	07-26-2022 04:14	a3e38c49
233st015_case_5.inp	547	07-19-2022 13:56	8ed57f0f
233st015_case_5.msg	1,490	07-26-2022 03:04	0ecff562
233st015_case_5.out	658,477	07-26-2022 03:38	1c9e5eeb
233st015_case_6.inp	550	07-19-2022 13:56	07148f3b
233st015_case_6.msg	1,490	07-26-2022 03:04	3e0e9532
233st015_case_6.out	658,863	07-26-2022 03:36	e9e6fa96
233st015_case_7.inp	755	07-19-2022 13:58	3400ec1e
233st015_case_7.msg	1,618	07-26-2022 03:04	669c952e
233st015_case_7.out	658,881	07-26-2022 03:47	f711e831
233st015_case_8.inp	550	07-19-2022 13:56	152b37de
233st015_case_8.msg	1,490	07-26-2022 03:04	73a6bc59
233st015_case_8.out	658,355	07-26-2022 03:35	47aacb1b
233st015_case_9.inp	550	07-19-2022 13:56	60ee27e8
233st015_case_9.msg	1,490	07-26-2022 03:04	06c3ce78
233st015_case_9.out	659,089	07-26-2022 03:34	0fcab4bd
233st015_case_10.inp	757	07-19-2022 13:58	802f7b11
233st015_case_10.msg	1,618	07-26-2022 03:04	365f0d82
233st015_case_10.out	659,845	07-26-2022 03:33	046312c4
233st015_case_11.inp	547	07-19-2022 13:56	88ea8637
233st015_case_11.msg	1,492	07-26-2022 03:04	c3d22ca6
233st015_case_11.out	658,032	07-26-2022 03:49	7e62853d
233st015_case_12.inp	547	07-19-2022 13:56	f00513d2
233st015_case_12.msg	1,492	07-26-2022 03:04	d1ce943e
233st015_case_12.out	658,505	07-26-2022 03:44	35c4ea77
233st015_case_13.inp	547	07-19-2022 13:56	6c99bd98
233st015_case_13.msg	1,492	07-26-2022 03:04	2751634b
233st015_case_13.out	657,747	07-26-2022 03:41	1547f993
233st015_case_14.inp	773	07-19-2022 13:58	da106ee8
233st015_case_14.msg	1,655	07-26-2022 03:04	df9f8303
233st015_case_14.out	661,670	07-26-2022 04:41	253f80a6
233st015_case_15.inp	604	07-19-2022 13:23	d80e7fee
233st015_case_15.msg	1,492	07-26-2022 03:04	f61f4a0e
233st015_case_15.out	658,738	07-26-2022 03:39	2f60eec7
233st015_case_16.inp	550	07-19-2022 13:56	fdc8e166
233st015_case_16.msg	1,492	07-26-2022 03:04	ea72691b
233st015_case_16.out	658,267	07-26-2022 03:39	e6e3434a
233st015_case_17.inp	755	07-19-2022 13:58	858396fa
233st015_case_17.msg	1,619	07-26-2022 03:04	7288ef34
233st015_case_17.out	660,272	07-26-2022 03:54	099148df
233st015_case_18.inp	543	07-19-2022 13:56	3b0887a1

CCO-CAL-0008 Rev 0 Attachment 1 Directories: 1, Files: 687, Size: 153,533,448 B			
Name	Size	Last write date	CRC32
233st015_case_18.msg	1,492	07-26-2022 03:04	a50bfd9b
233st015_case_18.out	658,355	07-26-2022 03:36	f8acc202
233st015_case_19.inp	550	07-19-2022 13:56	f8e5b74a
233st015_case_19.msg	1,492	07-26-2022 03:04	34db9dec
233st015_case_19.out	658,877	07-26-2022 03:35	2f3f30d8
233st015_case_20.inp	547	07-19-2022 13:56	cdab9fa5
233st015_case_20.msg	1,491	07-26-2022 03:04	cd1d793f
233st015_case_20.out	658,532	07-26-2022 03:55	b7e00a84
233st015_case_21.inp	547	07-19-2022 13:56	105cc615
233st015_case_21.msg	1,492	07-26-2022 03:04	71e5213a
233st015_case_21.out	657,301	07-26-2022 03:49	d458abda
233st015_case_22.inp	547	07-19-2022 13:56	7f29d7ab
233st015_case_22.msg	1,490	07-26-2022 03:04	1bd0e9d9
233st015_case_22.out	657,584	07-26-2022 03:43	a80c48ad
233st015_case_23.inp	547	07-19-2022 13:56	d5fef719
233st015_case_23.msg	1,492	07-26-2022 03:04	12c09a9f
233st015_case_23.out	657,912	07-26-2022 03:44	7b6f5e5d
233st015_case_24.inp	550	07-19-2022 13:56	375a4f97
233st015_case_24.msg	1,492	07-26-2022 03:04	d0993a26
233st015_case_24.out	657,617	07-26-2022 03:42	c263b7d5
233st015_case_25.inp	755	07-19-2022 13:58	bc57d88d
233st015_case_25.msg	1,620	07-26-2022 03:04	8b31425d
233st015_case_25.out	659,826	07-26-2022 04:21	80d3e98e
233st015_case_26.inp	547	07-19-2022 13:56	ad13e9db
233st015_case_26.msg	1,492	07-26-2022 03:04	ac253d78
233st015_case_26.out	658,161	07-26-2022 04:01	30ffcd1e
233st015_case_27.inp	545	07-19-2022 13:56	5c6683c8
233st015_case_27.msg	1,492	07-26-2022 03:04	514b0bb5
233st015_case_27.out	658,499	07-26-2022 03:56	835d788a
233st015_case_28.inp	550	07-19-2022 13:56	f95642b7
233st015_case_28.msg	1,492	07-26-2022 03:04	1fbc4672
233st015_case_28.out	657,756	07-26-2022 03:53	b96efa1a
233st015_case_29.inp	607	07-19-2022 13:23	085c964b
233st015_case_29.msg	1,492	07-26-2022 03:04	b4df96bd
233st015_case_29.out	659,112	07-26-2022 03:50	ccceba77
233st015_case_30.inp	607	07-19-2022 13:23	ebee4e9b
233st015_case_30.msg	1,492	07-26-2022 03:04	eb1a2ba3
233st015_case_30.out	658,925	07-26-2022 03:49	0b8bf658
233st015_case_31.inp	607	07-19-2022 13:23	70b9dd35
233st015_case_31.msg	1,492	07-26-2022 03:04	3e65d106
233st015_case_31.out	659,139	07-26-2022 03:47	90d32f0a
d4.inp	1,902	07-19-2022 14:13	37d5bbbd
d4.msg	1,662	07-26-2022 03:04	c2270aa0
d4.out	673,558	07-26-2022 05:23	eef6bb96
d4cs.inp	1,961	07-19-2022 14:13	bd007000
d4cs.msg	1,666	07-26-2022 03:04	f75b65f8
d4cs.out	675,007	07-26-2022 05:41	c9248355
pust001_case_1.inp	937	11-04-2019 19:43	7a0a09dd
pust001_case_1.msg	1,558	07-26-2022 03:05	4b339fe2
pust001_case_1.out	819,649	07-26-2022 06:17	8483a5f5
pust001_case_2.inp	936	11-04-2019 19:43	190e3e4f
pust001_case_2.msg	1,558	07-26-2022 03:05	33a77060
pust001_case_2.out	819,103	07-26-2022 06:11	18dc787d

CCO-CAL-0008 Rev 0 Attachment 1 Directories: 1, Files: 687, Size: 153,533,448 B			
Name	Size	Last write date	CRC32
pust001_case_3.inp	938	11-04-2019 19:43	4e6b0e79
pust001_case_3.msg	1,558	07-26-2022 03:05	bc93f0c2
pust001_case_3.out	819,327	07-26-2022 06:11	605e9197
pust001_case_4.inp	937	11-04-2019 19:43	fba56e0b
pust001_case_4.msg	1,558	07-26-2022 03:05	98db03fe
pust001_case_4.out	819,122	07-26-2022 06:10	66cb741b
pust001_case_5.inp	937	11-04-2019 19:43	21129e33
pust001_case_5.msg	1,558	07-26-2022 03:05	7ae5ddfa
pust001_case_5.out	819,303	07-26-2022 06:08	3002f66d
pust001_case_6.inp	943	11-04-2019 19:43	670df8d2
pust001_case_6.msg	1,558	07-26-2022 03:05	6d1cae24
pust001_case_6.out	819,799	07-26-2022 06:07	be0f7de1
pust002_case_1.inp	845	11-04-2019 19:43	6ea3b9f3
pust002_case_1.msg	1,418	07-26-2022 03:05	f2aab37c
pust002_case_1.out	818,788	07-26-2022 06:16	3e74a340
pust002_case_2.inp	845	11-04-2019 19:43	ac974019
pust002_case_2.msg	1,418	07-26-2022 03:05	dd332ce8
pust002_case_2.out	818,373	07-26-2022 06:01	b3e22dd8
pust002_case_3.inp	845	11-04-2019 19:43	cf0f71a0
pust002_case_3.msg	1,418	07-26-2022 03:05	69ee02c8
pust002_case_3.out	818,284	07-26-2022 06:15	55f922a2
pust002_case_4.inp	845	11-04-2019 19:43	797c0ebe
pust002_case_4.msg	1,417	07-26-2022 03:05	5827a122
pust002_case_4.out	817,607	07-26-2022 06:12	98e958e3
pust002_case_5.inp	851	11-04-2019 19:43	6271166c
pust002_case_5.msg	1,417	07-26-2022 03:05	89e4bde6
pust002_case_5.out	819,434	07-26-2022 06:12	125a1e9e
pust002_case_6.inp	851	11-04-2019 19:43	2d70bb34
pust002_case_6.msg	1,418	07-26-2022 03:05	7a26a444
pust002_case_6.out	819,014	07-26-2022 06:11	c4b99b90
pust002_case_7.inp	845	11-04-2019 19:43	d3f94b9f
pust002_case_7.msg	1,418	07-26-2022 03:05	a73b2c68
pust002_case_7.out	817,127	07-26-2022 06:08	21456d2b
pust003_case_1.inp	1,004	11-04-2019 19:43	f309be10
pust003_case_1.msg	1,418	07-26-2022 03:05	ea5b3a12
pust003_case_1.out	819,842	07-26-2022 06:13	eb0c31f3
pust003_case_2.inp	1,004	11-04-2019 19:43	b84a7c03
pust003_case_2.msg	1,418	07-26-2022 03:05	1dc69674
pust003_case_2.out	820,181	07-26-2022 06:17	2d009117
pust003_case_3.inp	1,004	11-04-2019 19:43	5085b459
pust003_case_3.msg	1,418	07-26-2022 03:05	c68c4074
pust003_case_3.out	819,684	07-26-2022 06:13	5ec21d57
pust003_case_4.inp	1,004	11-04-2019 19:43	16803b62
pust003_case_4.msg	1,418	07-26-2022 03:05	e19c94fd
pust003_case_4.out	819,606	07-26-2022 06:17	3928ecbc
pust003_case_5.inp	1,004	11-04-2019 19:43	2bb7772a
pust003_case_5.msg	1,418	07-26-2022 03:05	1650efb5
pust003_case_5.out	820,083	07-26-2022 06:11	10687503
pust003_case_6.inp	1,004	11-04-2019 19:43	8a037d6b
pust003_case_6.msg	1,418	07-26-2022 03:05	0f7140fb
pust003_case_6.out	819,906	07-26-2022 06:12	f6daaf2b
pust003_case_7.inp	1,004	11-04-2019 19:43	494c0135
pust003_case_7.msg	1,347	07-26-2022 03:05	49dfb823
pust003_case_7.out	818,545	07-26-2022 06:18	3c696451
pust003_case_8.inp	1,004	11-04-2019 19:43	8b86ca29
pust003_case_8.msg	1,348	07-26-2022 03:05	2b0c6169
pust003_case_8.out	818,812	07-26-2022 06:16	e2b9e508
pust004_case_1.inp	855	11-04-2019 19:43	844048f9
pust004_case_1.msg	1,418	07-26-2022 03:05	c122e3d9

CCO-CAL-0008 Rev 0 Attachment 1 Directories: 1, Files: 687, Size: 153,533,448 B			
Name	Size	Last write date	CRC32
pust004_case_1.out	819,101	07-26-2022 06:17	ea5da5e3
pust004_case_2.inp	852	11-04-2019 19:43	8333ddfc
pust004_case_2.msg	1,418	07-26-2022 03:05	1b5c40ef
pust004_case_2.out	818,386	07-26-2022 06:16	7d076f26
pust004_case_3.inp	851	11-04-2019 19:43	580ed1b5
pust004_case_3.msg	1,418	07-26-2022 03:05	19f4e821
pust004_case_3.out	818,471	07-26-2022 06:16	fbf8e32e
pust004_case_4.inp	848	11-04-2019 19:43	93a9341f
pust004_case_4.msg	1,418	07-26-2022 03:05	2fce5e1a
pust004_case_4.out	818,791	07-26-2022 06:14	3ea15785
pust004_case_5.inp	851	11-04-2019 19:43	e3745cf5
pust004_case_5.msg	1,418	07-26-2022 03:05	bc7cdac6
pust004_case_5.out	817,955	07-26-2022 06:16	9ecb00e4
pust004_case_6.inp	851	11-04-2019 19:43	e5e3569a
pust004_case_6.msg	1,418	07-26-2022 03:05	76cf6749
pust004_case_6.out	818,143	07-26-2022 06:17	42d35d7f
pust004_case_7.inp	851	11-04-2019 19:43	5a3f7338
pust004_case_7.msg	1,418	07-26-2022 03:05	5045020a
pust004_case_7.out	818,392	07-26-2022 06:10	a2015d28
pust004_case_8.inp	851	11-04-2019 19:43	5b940060
pust004_case_8.msg	1,418	07-26-2022 03:05	f490a0a7
pust004_case_8.out	819,876	07-26-2022 06:13	86197a0a
pust004_case_9.inp	851	11-04-2019 19:43	d46106ff
pust004_case_9.msg	1,418	07-26-2022 03:05	b9a22767
pust004_case_9.out	818,961	07-26-2022 06:10	33139de7
pust004_case_10.inp	853	11-04-2019 19:43	4b2aa2de
pust004_case_10.msg	1,420	07-26-2022 03:05	5caec402
pust004_case_10.out	818,541	07-26-2022 06:17	79fd6d20
pust004_case_11.inp	853	11-04-2019 19:43	8887f913
pust004_case_11.msg	1,420	07-26-2022 03:05	a1b44cee
pust004_case_11.out	819,119	07-26-2022 06:13	7b54add1
pust004_case_12.inp	853	11-04-2019 19:43	6620225f
pust004_case_12.msg	1,420	07-26-2022 03:05	d3733098
pust004_case_12.out	819,469	07-26-2022 06:19	bfd6f5d0
pust004_case_13.inp	853	11-04-2019 19:43	345bdfe7
pust004_case_13.msg	1,420	07-26-2022 03:05	3d1226b2
pust004_case_13.out	817,773	07-26-2022 06:15	b18f3ea5
pust005_case_1.inp	847	11-04-2019 19:43	fefc618f
pust005_case_1.msg	1,418	07-26-2022 03:05	208330b8
pust005_case_1.out	819,196	07-26-2022 06:12	40f13974
pust005_case_2.inp	849	11-04-2019 19:43	b93f7476
pust005_case_2.msg	1,418	07-26-2022 03:05	dc8b6516
pust005_case_2.out	818,171	07-26-2022 06:11	c7eff612
pust005_case_3.inp	845	11-04-2019 19:43	c6451d9f
pust005_case_3.msg	1,418	07-26-2022 03:05	31eedf48
pust005_case_3.out	818,459	07-26-2022 06:11	bac5168f
pust005_case_4.inp	848	11-04-2019 19:43	4d9c5412
pust005_case_4.msg	1,418	07-26-2022 03:05	5b91ad7b
pust005_case_4.out	819,010	07-26-2022 06:10	bffe51c7
pust005_case_5.inp	845	11-04-2019 19:43	75776cc8
pust005_case_5.msg	1,418	07-26-2022 03:05	a7d80208
pust005_case_5.out	818,354	07-26-2022 06:10	4d66b836
pust005_case_6.inp	845	11-04-2019 19:43	00ef4e51
pust005_case_6.msg	1,418	07-26-2022 03:05	ba037da9
pust005_case_6.out	818,478	07-26-2022 06:07	3bed51cf
pust005_case_7.inp	845	11-04-2019 19:43	05359628
pust005_case_7.msg	1,418	07-26-2022 03:05	6cb3d822
pust005_case_7.out	818,688	07-26-2022 06:09	b64e356a
pust005_case_8.inp	845	11-04-2019 19:43	17a0fba0

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Name	Size	Last write date	CRC32
pust005_case_8.msg	1,418	07-26-2022 03:05	d34c3c01
pust005_case_8.out	818,996	07-26-2022 06:11	56a66a87
pust005_case_9.inp	845	11-04-2019 19:43	f5369cce
pust005_case_9.msg	1,418	07-26-2022 03:05	7020ce53
pust005_case_9.out	819,310	07-26-2022 06:11	3baac62d
pust006_case_1.inp	942	11-04-2019 19:43	ebe029c5
pust006_case_1.msg	1,418	07-26-2022 03:05	df1e681d
pust006_case_1.out	660,215	07-26-2022 06:13	c45085f1
pust006_case_2.inp	921	11-04-2019 19:43	d1e10c5e
pust006_case_2.msg	1,418	07-26-2022 03:05	c927e82c
pust006_case_2.out	660,277	07-26-2022 06:12	d869d425
pust006_case_3.inp	901	11-04-2019 19:43	b8a61672
pust006_case_3.msg	1,418	07-26-2022 03:05	7834d7c5
pust006_case_3.out	660,304	07-26-2022 06:11	445271c0
pust007_case_2.inp	986	11-04-2019 19:43	8b691209
pust007_case_2.msg	1,558	07-26-2022 03:05	d8352c58
pust007_case_2.out	858,782	07-26-2022 06:13	25478cc0
pust007_case_3.inp	986	11-04-2019 19:43	e5bfc58f
pust007_case_3.msg	1,558	07-26-2022 03:05	51c0ba3b
pust007_case_3.out	857,499	07-26-2022 06:16	535b687c
pust007_case_5.inp	985	11-04-2019 19:43	f859120e
pust007_case_5.msg	1,558	07-26-2022 03:05	795545a5
pust007_case_5.out	858,893	07-26-2022 06:10	5c911cf1
pust007_case_6.inp	985	11-04-2019 19:43	6ee7113c
pust007_case_6.msg	1,558	07-26-2022 03:05	8c557c63
pust007_case_6.out	857,239	07-26-2022 06:12	ba195db3
pust007_case_7.inp	985	11-04-2019 19:43	1d9a73a0
pust007_case_7.msg	1,558	07-26-2022 03:05	9790462c
pust007_case_7.out	857,334	07-26-2022 06:12	cead7bf1
pust007_case_8.inp	985	11-04-2019 19:43	2b5e6e71
pust007_case_8.msg	1,558	07-26-2022 03:05	7a6a1e68
pust007_case_8.out	858,341	07-26-2022 06:10	e45e37ba
pust007_case_9.inp	985	11-04-2019 19:43	ff5a91bc
pust007_case_9.msg	1,558	07-26-2022 03:05	ee30be85
pust007_case_9.out	858,245	07-26-2022 06:13	bf2d14f3
pust007_case_10.inp	985	11-04-2019 19:43	1ba7351d
pust007_case_10.msg	1,560	07-26-2022 03:05	8a094b26
pust007_case_10.out	858,527	07-26-2022 06:14	40e2cc95
pust009_case_1.inp	1,467	11-04-2019 19:43	7ac950b0
pust009_case_1.msg	3,098	07-26-2022 03:05	046c10c1
pust009_case_1.out	842,618	07-26-2022 10:04	bbf3a8fa
pust009_case_2.inp	1,467	11-04-2019 19:43	7491a55e
pust009_case_2.msg	3,098	07-26-2022 03:05	9a8b6757
pust009_case_2.out	842,620	07-26-2022 10:21	32f0ed86
pust009_case_3.inp	1,422	11-04-2019 19:43	68ee86ea
pust009_case_3.msg	3,098	07-26-2022 03:05	0ef28e8a
pust009_case_3.out	804,462	07-26-2022 10:25	a52a8fb0
pust010_case_1.9.inp	1,390	11-04-2019 19:43	f35159cf
pust010_case_1.9.msg	1,422	07-26-2022 03:05	1235e097
pust010_case_1.9.out	664,383	07-26-2022 06:10	48faf42f
pust010_case_1.11.inp	1,401	11-04-2019 19:43	7dbb525a
pust010_case_1.11.msg	1,424	07-26-2022 03:05	526577b0
pust010_case_1.11.out	664,167	07-26-2022 06:06	d70adcc4
pust010_case_1.12.inp	1,398	11-04-2019 19:43	c51efcff
pust010_case_1.12.msg	1,424	07-26-2022 03:05	77f0001a
pust010_case_1.12.out	664,635	07-26-2022 06:10	c34f3027
pust010_case_2.9.inp	1,391	11-04-2019 19:43	d04423dc
pust010_case_2.9.msg	1,422	07-26-2022 03:05	0cdf0794
pust010_case_2.9.out	663,114	07-26-2022 06:19	c03191da

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Name	Size	Last write date	CRC32
pust010_case_2.11.inp	1,403	11-04-2019 19:43	129839f2
pust010_case_2.11.msg	1,424	07-26-2022 03:05	2fdc7ca3
pust010_case_2.11.out	664,382	07-26-2022 06:05	0feaa7c2
pust010_case_2.12.inp	1,400	11-04-2019 19:43	d881e172
pust010_case_2.12.msg	1,424	07-26-2022 03:05	6598f0db
pust010_case_2.12.out	664,958	07-26-2022 06:10	9112feb9
pust010_case_3.9.inp	1,399	11-04-2019 19:43	1d3794af
pust010_case_3.9.msg	1,422	07-26-2022 03:27	0439b5bf
pust010_case_3.9.out	664,131	07-26-2022 06:33	daf442d3
pust010_case_3.11.inp	1,386	11-04-2019 19:43	ee25dc02
pust010_case_3.11.msg	1,424	07-26-2022 03:27	6fc9859c
pust010_case_3.11.out	663,733	07-26-2022 06:33	ff4697e8
pust010_case_3.12.inp	1,398	11-04-2019 19:43	2f3f986e
pust010_case_3.12.msg	1,424	07-26-2022 03:27	bae45605
pust010_case_3.12.out	664,475	07-26-2022 06:33	429d7203
pust010_case_4.11.inp	1,397	11-04-2019 19:43	6c82e546
pust010_case_4.11.msg	1,424	07-26-2022 03:27	cac55454
pust010_case_4.11.out	664,431	07-26-2022 06:32	5b6f6a53
pust010_case_4.12.inp	1,400	11-04-2019 19:43	92bc7426
pust010_case_4.12.msg	1,424	07-26-2022 03:27	87a2767d
pust010_case_4.12.out	664,439	07-26-2022 06:33	e2f7888a
pust010_case_5.11.inp	1,400	11-04-2019 19:43	d6245c76
pust010_case_5.11.msg	1,424	07-26-2022 03:27	ace303c2
pust010_case_5.11.out	664,146	07-26-2022 06:32	2218a0ac
pust010_case_6.11.inp	1,403	11-04-2019 19:43	b8c894aa
pust010_case_6.11.msg	1,424	07-26-2022 03:27	fb27e704
pust010_case_6.11.out	663,535	07-26-2022 06:32	ddf31d07
pust010_case_7.11.inp	1,401	11-04-2019 19:43	79067971
pust010_case_7.11.msg	1,424	07-26-2022 03:27	6848dcc5
pust010_case_7.11.out	664,352	07-26-2022 06:33	9753d4a9
pust011_case_1.16.inp	774	11-04-2019 19:43	90408899
pust011_case_1.16.msg	1,424	07-26-2022 03:27	309137ed
pust011_case_1.16.out	657,900	07-26-2022 04:23	fe609328
pust011_case_1.18.inp	902	11-04-2019 19:43	24c1bb1d
pust011_case_1.18.msg	1,704	07-26-2022 03:27	0df17b93
pust011_case_1.18.out	663,783	07-26-2022 04:47	7ed3c069
pust011_case_2.16.inp	774	11-04-2019 19:43	e6f0eec9
pust011_case_2.16.msg	1,424	07-26-2022 03:27	1743e5bc
pust011_case_2.16.out	657,262	07-26-2022 04:22	7c6e3ab1
pust011_case_2.18.inp	902	11-04-2019 19:43	160d1b91
pust011_case_2.18.msg	1,704	07-26-2022 03:27	3df6ab65
pust011_case_2.18.out	662,975	07-26-2022 04:44	b88c0c08
pust011_case_3.16.inp	774	11-04-2019 19:43	ca3a21b1
pust011_case_3.16.msg	1,424	07-26-2022 03:27	e79a32f9
pust011_case_3.16.out	658,568	07-26-2022 04:25	5f11bd16
pust011_case_3.18.inp	904	11-04-2019 19:43	76d5fb18
pust011_case_3.18.msg	1,704	07-26-2022 03:27	00580f17
pust011_case_3.18.out	662,555	07-26-2022 04:50	fdacc60c
pust011_case_4.16.inp	774	11-04-2019 19:43	a61785c5
pust011_case_4.16.msg	1,424	07-26-2022 03:27	4e820e76
pust011_case_4.16.out	657,511	07-26-2022 04:24	abef9258
pust011_case_4.18.inp	901	11-04-2019 19:43	d2b8d3fc
pust011_case_4.18.msg	1,704	07-26-2022 03:27	7f0a83b6
pust011_case_4.18.out	663,789	07-26-2022 04:47	e65e9166
pust011_case_5.16.inp	774	11-04-2019 19:43	1647cb37
pust011_case_5.16.msg	1,424	07-26-2022 03:27	b7d9734d
pust011_case_5.16.out	658,054	07-26-2022 04:19	f8eee65f
pust011_case_5.18.inp	901	11-04-2019 19:43	54f9cb48
pust011_case_5.18.msg	1,704	07-26-2022 03:27	3f5c2c36

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Name	Size	Last write date	CRC32
pust011_case_5.18.out	663,468	07-26-2022 04:45	da56b60b
pust011_case_6.18.inp	901	11-04-2019 19:43	68e27933
pust011_case_6.18.msg	1,704	07-26-2022 03:27	619edcf0
pust011_case_6.18.out	663,583	07-26-2022 04:38	9c40d968
pust011_case_7.18.inp	902	11-04-2019 19:43	ab9e9fca
pust011_case_7.18.msg	1,704	07-26-2022 03:27	bc0f081e
pust011_case_7.18.out	662,926	07-26-2022 04:44	bb1efc3c
pust014_case_1.inp	2,539	11-04-2019 19:43	7622ef5a
pust014_case_1.msg	2,188	07-26-2022 03:27	93c1bb49
pust014_case_1.out	518,547	07-26-2022 06:29	46017f6d
pust014_case_3.inp	2,541	11-04-2019 19:43	6df0d268
pust014_case_3.msg	2,188	07-26-2022 03:28	fd599830
pust014_case_3.out	518,509	07-26-2022 06:58	eb0e7173
pust014_case_4.inp	2,570	11-04-2019 19:43	3916c581
pust014_case_4.msg	2,188	07-26-2022 03:28	63eefd14
pust014_case_4.out	519,769	07-26-2022 06:56	326746b4
pust014_case_5.inp	2,541	11-04-2019 19:43	36827dc0
pust014_case_5.msg	2,188	07-26-2022 03:28	915bf15f
pust014_case_5.out	519,350	07-26-2022 06:56	aa415aed
pust014_case_6.inp	2,542	11-04-2019 19:43	485cdd73
pust014_case_6.msg	2,188	07-26-2022 03:30	2b989f4e
pust014_case_6.out	678,278	07-26-2022 07:04	9b537033
pust014_case_7.inp	2,547	11-04-2019 19:43	e8e32cf3
pust014_case_7.msg	2,188	07-26-2022 03:30	c02fc43a
pust014_case_7.out	518,396	07-26-2022 07:04	19506a03
pust014_case_8.inp	2,519	11-04-2019 19:43	42f9bd71
pust014_case_8.msg	2,188	07-26-2022 03:30	6f63bddb
pust014_case_8.out	519,060	07-26-2022 07:03	0984e398
pust014_case_9.inp	2,520	11-04-2019 19:43	7e0de4a7
pust014_case_9.msg	2,188	07-26-2022 03:30	bf9f62a6
pust014_case_9.out	518,355	07-26-2022 07:02	7447a8bd
pust014_case_10.inp	2,520	11-04-2019 19:43	c9f63ef5
pust014_case_10.msg	2,190	07-26-2022 03:27	87b36339
pust014_case_10.out	519,429	07-26-2022 06:27	7747dde5
pust014_case_11.inp	2,521	11-04-2019 19:43	7f87c80a
pust014_case_11.msg	2,190	07-26-2022 03:27	72678265
pust014_case_11.out	678,409	07-26-2022 06:28	70e16aec
pust014_case_12.inp	2,524	11-04-2019 19:43	b57149ad
pust014_case_12.msg	2,190	07-26-2022 03:27	3b4d8e4c
pust014_case_12.out	677,448	07-26-2022 06:28	ec7e56d1
pust014_case_13.inp	2,534	11-04-2019 19:43	ac98e437
pust014_case_13.msg	2,190	07-26-2022 03:28	f33d92a0
pust014_case_13.out	519,069	07-26-2022 06:59	13350ac1
pust014_case_14.inp	2,528	11-04-2019 19:43	5293708d
pust014_case_14.msg	2,190	07-26-2022 03:28	8cb88c3c
pust014_case_14.out	518,885	07-26-2022 06:57	e07f0e98
pust014_case_15.inp	2,529	11-04-2019 19:43	55799319
pust014_case_15.msg	2,190	07-26-2022 03:28	90521945
pust014_case_15.out	519,189	07-26-2022 06:57	00abb74e
pust014_case_16.inp	2,529	11-04-2019 19:43	439e1653
pust014_case_16.msg	2,190	07-26-2022 03:28	ab9d3945
pust014_case_16.out	518,132	07-26-2022 06:56	6a58a712
pust014_case_17.inp	2,530	11-04-2019 19:43	26096a86
pust014_case_17.msg	2,190	07-26-2022 03:28	3003b48b
pust014_case_17.out	677,591	07-26-2022 06:53	4fdfea0f
pust014_case_18.inp	2,579	11-04-2019 19:43	b11b029a
pust014_case_18.msg	2,190	07-26-2022 03:28	27c51677
pust014_case_18.out	518,634	07-26-2022 06:56	6ea4a900
pust014_case_19.inp	2,569	11-04-2019 19:43	a900349a

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Name	Size	Last write date	CRC32
pust014_case_19.msg	2,190	07-26-2022 03:28	000f8476
pust014_case_19.out	519,284	07-26-2022 06:56	15cc44d5
pust014_case_20.inp	2,568	11-04-2019 19:43	7ec4fe02
pust014_case_20.msg	2,190	07-26-2022 03:28	8fff7958
pust014_case_20.out	518,763	07-26-2022 06:56	1841b0d3
pust014_case_21.inp	2,570	11-04-2019 19:43	33f37495
pust014_case_21.msg	2,190	07-26-2022 03:28	79d9ee9a
pust014_case_21.out	519,704	07-26-2022 06:52	7cc59ec9
pust014_case_22.inp	2,572	11-04-2019 19:43	82f9092f
pust014_case_22.msg	2,190	07-26-2022 03:28	0f28b6b9
pust014_case_22.out	678,222	07-26-2022 06:53	bccb11ff
pust014_case_23.inp	2,574	11-04-2019 19:43	664ac614
pust014_case_23.msg	2,190	07-26-2022 03:28	790ec38d
pust014_case_23.out	678,439	07-26-2022 06:53	0de9f3af
pust014_case_24.inp	2,632	11-04-2019 19:43	b4ef0101
pust014_case_24.msg	2,190	07-26-2022 03:28	7d99071a
pust014_case_24.out	519,180	07-26-2022 06:54	b1b71c32
pust014_case_25.inp	2,653	11-04-2019 19:43	5c842227
pust014_case_25.msg	2,190	07-26-2022 03:28	398c1e85
pust014_case_25.out	519,708	07-26-2022 06:57	e0d34779
pust014_case_26.inp	2,623	11-04-2019 19:43	7a78a932
pust014_case_26.msg	2,190	07-26-2022 03:28	112c3c93
pust014_case_26.out	518,630	07-26-2022 06:56	700b3886
pust014_case_27.inp	2,623	11-04-2019 19:43	859abe4a
pust014_case_27.msg	2,190	07-26-2022 03:28	5d88976a
pust014_case_27.out	520,114	07-26-2022 06:56	b6baf928
pust014_case_28.inp	2,625	11-04-2019 19:43	1a4ea4d5
pust014_case_28.msg	2,190	07-26-2022 03:28	fdc0c21e
pust014_case_28.out	678,567	07-26-2022 06:56	307509fe
pust014_case_29.inp	2,624	11-04-2019 19:43	2550f414
pust014_case_29.msg	2,190	07-26-2022 03:28	95f4f54c
pust014_case_29.out	678,679	07-26-2022 06:56	62b367eb
pust014_case_30.inp	2,669	11-04-2019 19:43	777227e9
pust014_case_30.msg	2,190	07-26-2022 03:28	18bb381e
pust014_case_30.out	518,581	07-26-2022 06:57	fed5d607
pust014_case_31.inp	2,662	11-04-2019 19:43	b7fce2dc
pust014_case_31.msg	2,190	07-26-2022 03:28	302412b9
pust014_case_31.out	518,968	07-26-2022 06:56	c15e9bdb
pust014_case_33.inp	2,631	11-04-2019 19:43	197c926d
pust014_case_33.msg	2,190	07-26-2022 03:28	0e7e9150
pust014_case_33.out	519,503	07-26-2022 06:53	07426875
pust014_case_34.inp	2,631	11-04-2019 19:43	2659679f
pust014_case_34.msg	2,190	07-26-2022 03:28	74c6f3f3
pust014_case_34.out	519,756	07-26-2022 06:53	8e228eb6
pust015_case_1.inp	2,572	11-04-2019 19:43	e449af84
pust015_case_1.msg	2,188	07-26-2022 03:30	46df775d
pust015_case_1.out	518,566	07-26-2022 07:00	8496495f
pust015_case_2.inp	2,572	11-04-2019 19:43	3b169d3e
pust015_case_2.msg	2,188	07-26-2022 03:30	9485873f
pust015_case_2.out	518,817	07-26-2022 06:49	a2c80245
pust015_case_3.inp	2,573	11-04-2019 19:43	8fb49903
pust015_case_3.msg	2,188	07-26-2022 03:30	8f48c3d9
pust015_case_3.out	518,703	07-26-2022 06:49	ba064eb5
pust015_case_4.inp	2,573	11-04-2019 19:43	9ec6efa7
pust015_case_4.msg	2,188	07-26-2022 03:30	290b061f
pust015_case_4.out	519,123	07-26-2022 06:48	8bf66059
pust015_case_5.inp	2,573	11-04-2019 19:43	9ddc8963
pust015_case_5.msg	2,188	07-26-2022 03:30	3275ae34
pust015_case_5.out	519,418	07-26-2022 06:49	9c8845ce

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Name	Size	Last write date	CRC32
pust015_case_6.inp	2,524	11-04-2019 19:43	5b5176df
pust015_case_6.msg	2,188	07-26-2022 03:30	40bb11e2
pust015_case_6.out	519,149	07-26-2022 06:51	6cf748eb
pust015_case_7.inp	2,687	11-04-2019 19:43	f9aa24f7
pust015_case_7.msg	2,188	07-26-2022 03:30	06ae1514
pust015_case_7.out	519,011	07-26-2022 06:49	0f0333c0
pust015_case_8.inp	2,688	11-04-2019 19:43	c57da314
pust015_case_8.msg	2,188	07-26-2022 03:30	50f0ad12
pust015_case_8.out	519,141	07-26-2022 06:48	9d6900d0
pust015_case_9.inp	2,688	11-04-2019 19:43	45e6bd24
pust015_case_9.msg	2,188	07-26-2022 03:30	fb248f3f
pust015_case_9.out	519,268	07-26-2022 06:11	e00e721c
pust015_case_10.inp	2,656	11-04-2019 19:43	5197b7df
pust015_case_10.msg	2,190	07-26-2022 03:30	74a431d0
pust015_case_10.out	519,577	07-26-2022 06:58	fe46695e
pust015_case_11.inp	2,655	11-04-2019 19:43	a0ce2f0e
pust015_case_11.msg	2,190	07-26-2022 03:30	bad00093
pust015_case_11.out	519,371	07-26-2022 07:01	59b8fcca
pust015_case_12.inp	2,656	11-04-2019 19:43	11ff243e
pust015_case_12.msg	2,190	07-26-2022 03:30	b4298fce
pust015_case_12.out	518,913	07-26-2022 07:00	fc2c7350
pust015_case_13.inp	2,656	11-04-2019 19:43	0f0219b6
pust015_case_13.msg	2,190	07-26-2022 03:30	d37f1ddb
pust015_case_13.out	519,526	07-26-2022 06:54	eab6cca1
pust015_case_14.inp	2,656	11-04-2019 19:43	c97b8e08
pust015_case_14.msg	2,190	07-26-2022 03:30	deddf409
pust015_case_14.out	519,667	07-26-2022 06:54	0274f98e
pust015_case_15.inp	2,674	11-04-2019 19:43	d2b54830
pust015_case_15.msg	2,190	07-26-2022 03:30	167351c6
pust015_case_15.out	519,460	07-26-2022 06:59	429de561
pust015_case_16.inp	2,663	11-04-2019 19:43	22e2c078
pust015_case_16.msg	2,190	07-26-2022 03:30	366b36aa
pust015_case_16.out	519,173	07-26-2022 06:57	f3652793
pust015_case_17.inp	2,665	11-04-2019 19:43	70cbcbbf
pust015_case_17.msg	2,190	07-26-2022 03:30	294acabf
pust015_case_17.out	519,652	07-26-2022 06:47	70f980b4
pust016_case_1.inp	2,344	11-04-2019 19:43	7d5986a0
pust016_case_1.msg	2,117	07-26-2022 03:30	f36c0b7f
pust016_case_1.out	513,357	07-26-2022 05:09	073a8fa6
pust016_case_2.inp	2,343	11-04-2019 19:43	c8782e6b
pust016_case_2.msg	2,117	07-26-2022 03:30	439ece8f
pust016_case_2.out	513,917	07-26-2022 05:24	67b99d27
pust016_case_3.inp	2,343	11-04-2019 19:43	4289b998
pust016_case_3.msg	2,118	07-26-2022 03:30	1a599672
pust016_case_3.out	514,578	07-26-2022 05:24	628986ee
pust016_case_4.inp	2,343	11-04-2019 19:43	1e3fd929
pust016_case_4.msg	2,118	07-26-2022 03:30	cce0301a
pust016_case_4.out	514,158	07-26-2022 05:24	08a76b27
pust016_case_5.inp	2,332	11-04-2019 19:43	cadf86b5
pust016_case_5.msg	2,118	07-26-2022 03:30	77fc7d9c
pust016_case_5.out	514,055	07-26-2022 05:27	d77d7bd8
pust016_case_6.inp	2,331	11-04-2019 19:43	9b87e205
pust016_case_6.msg	2,118	07-26-2022 03:30	4e66f1eb
pust016_case_6.out	513,638	07-26-2022 05:27	cf8ce7ad
pust016_case_7.inp	2,331	11-04-2019 19:43	9c9ea309
pust016_case_7.msg	2,118	07-26-2022 03:30	2a8b4f2d
pust016_case_7.out	514,631	07-26-2022 05:27	d4f57462
pust016_case_8.inp	2,331	11-04-2019 19:43	f42b22ef
pust016_case_8.msg	2,118	07-26-2022 03:30	cba18943

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Name	Size	Last write date	CRC32
pust016_case_8.out	513,306	07-26-2022 05:25	a22a9611
pust016_case_9.inp	2,279	11-04-2019 19:43	9d73ac49
pust016_case_9.msg	1,768	07-26-2022 03:30	b9e13d9e
pust016_case_9.out	510,679	07-26-2022 05:30	635d4baf
pust016_case_10.inp	2,278	11-04-2019 19:43	e9bb5b8f
pust016_case_10.msg	1,770	07-26-2022 03:30	36b04497
pust016_case_10.out	510,213	07-26-2022 05:18	5506a2f5
pust016_case_11.inp	2,278	11-04-2019 19:43	84bab108
pust016_case_11.msg	1,770	07-26-2022 03:30	84de03b7
pust016_case_11.out	511,401	07-26-2022 05:16	d2ae88e0
pust017_case_1.inp	2,325	11-04-2019 19:43	2e56b29c
pust017_case_1.msg	2,118	07-26-2022 03:30	5982adca
pust017_case_1.out	512,904	07-26-2022 05:28	09f60c9a
pust017_case_2.inp	2,369	11-04-2019 19:43	a458ad75
pust017_case_2.msg	2,118	07-26-2022 03:31	cfacbf94
pust017_case_2.out	514,090	07-26-2022 05:28	b64ac3cc
pust017_case_3.inp	2,369	11-04-2019 19:43	2520d1e4
pust017_case_3.msg	2,118	07-26-2022 03:30	2b2baaee
pust017_case_3.out	513,267	07-26-2022 05:27	7dc1e35b
pust017_case_4.inp	2,369	11-04-2019 19:43	05d431d0
pust017_case_4.msg	2,118	07-26-2022 03:30	5555441e
pust017_case_4.out	513,310	07-26-2022 05:25	6b75220c
pust017_case_5.inp	2,369	11-04-2019 19:43	f22c5373
pust017_case_5.msg	2,118	07-26-2022 03:30	5025b3e8
pust017_case_5.out	513,217	07-26-2022 05:25	027f3eda
pust017_case_6.inp	2,369	11-04-2019 19:43	480a89c8
pust017_case_6.msg	2,118	07-26-2022 03:30	9b03cec8
pust017_case_6.out	513,871	07-26-2022 05:26	116a2fff
pust017_case_7.inp	2,369	11-04-2019 19:43	eb7e66f5
pust017_case_7.msg	2,117	07-26-2022 03:30	e368599b
pust017_case_7.out	514,290	07-26-2022 05:24	b2fa8dda
pust017_case_8.inp	2,366	11-04-2019 19:43	b3769abf
pust017_case_8.msg	2,118	07-26-2022 03:34	91421c57
pust017_case_8.out	514,900	07-26-2022 04:58	653177f6
pust017_case_9.inp	2,368	11-04-2019 19:43	bb170491
pust017_case_9.msg	2,118	07-26-2022 03:35	661b410a
pust017_case_9.out	514,197	07-26-2022 04:56	e5f946f7
pust017_case_10.inp	2,368	11-04-2019 19:43	d5e15431
pust017_case_10.msg	2,120	07-26-2022 03:30	011da38a
pust017_case_10.out	513,871	07-26-2022 05:27	a091ae67
pust017_case_11.inp	2,293	11-04-2019 19:43	6f7cd2dc
pust017_case_11.msg	2,120	07-26-2022 03:30	1724e0b9
pust017_case_11.out	513,416	07-26-2022 05:27	bbd92b01
pust017_case_12.inp	2,369	11-04-2019 19:43	829038f4
pust017_case_12.msg	2,120	07-26-2022 03:30	76fc34e4
pust017_case_12.out	514,219	07-26-2022 05:27	f2bc501a
pust017_case_13.inp	2,371	11-04-2019 19:43	f3dfc276
pust017_case_13.msg	2,120	07-26-2022 03:30	37c2beb3
pust017_case_13.out	514,373	07-26-2022 05:29	32e226b3
pust017_case_14.inp	2,369	11-04-2019 19:43	af1d0e53
pust017_case_14.msg	2,120	07-26-2022 03:30	4702cd11
pust017_case_14.out	513,707	07-26-2022 05:27	87901d18
pust017_case_15.inp	2,369	11-04-2019 19:43	1967df61
pust017_case_15.msg	2,120	07-26-2022 03:30	d3862b4b
pust017_case_15.out	514,207	07-26-2022 05:29	1faad2bc
pust017_case_16.inp	2,404	11-04-2019 19:43	a831a1c0
pust017_case_16.msg	2,120	07-26-2022 03:30	c0f45e0b
pust017_case_16.out	514,220	07-26-2022 05:29	5bb9cdc4
pust017_case_17.inp	2,371	11-04-2019 19:43	1233c488

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Name	Size	Last write date	CRC32
pust017_case_17.msg	2,120	07-26-2022 03:30	85811760
pust017_case_17.out	513,685	07-26-2022 05:29	6a02c9e8
pust017_case_18.inp	2,404	11-04-2019 19:43	7b2168a9
pust017_case_18.msg	2,120	07-26-2022 03:30	fdfd1f50
pust017_case_18.out	513,885	07-26-2022 05:28	cbc6f04e
pust020_case_1.inp	1,005	11-04-2019 19:43	374d4747
pust020_case_1.msg	1,558	07-26-2022 03:36	bdbddc43
pust020_case_1.out	858,462	07-26-2022 05:57	a09b154d
pust020_case_2.inp	1,006	11-04-2019 19:43	c1c4b5fa
pust020_case_2.msg	1,558	07-26-2022 03:36	f643e307
pust020_case_2.out	858,416	07-26-2022 05:55	96d1c9d4
pust020_case_3.inp	1,005	11-04-2019 19:43	40cbe04e
pust020_case_3.msg	1,558	07-26-2022 03:37	35b2f76d
pust020_case_3.out	858,569	07-26-2022 06:04	d3622a79
pust020_case_5.inp	1,005	11-04-2019 19:43	7afabb1d
pust020_case_5.msg	1,557	07-26-2022 03:37	46593e92
pust020_case_5.out	858,551	07-26-2022 06:05	224e2377
pust020_case_6.inp	1,005	11-04-2019 19:43	84540964
pust020_case_6.msg	1,558	07-26-2022 03:39	7f0e03d9
pust020_case_6.out	857,253	07-26-2022 06:02	021b6591
pust020_case_7.inp	1,005	11-04-2019 19:43	2f99eb34
pust020_case_7.msg	1,558	07-26-2022 03:39	387bc841
pust020_case_7.out	858,804	07-26-2022 05:59	873eba32
pust020_case_8.inp	1,079	11-04-2019 19:43	8d1e5043
pust020_case_8.msg	1,838	07-26-2022 03:40	a7a06b8a
pust020_case_8.out	901,551	07-26-2022 05:34	4e41e59f
pust020_case_9.inp	1,079	11-04-2019 19:43	35e1564a
pust020_case_9.msg	1,838	07-26-2022 03:41	8adf9a31
pust020_case_9.out	900,547	07-26-2022 05:42	a08d2f65
pust021_case_7.inp	589	11-04-2019 19:43	7160d775
pust021_case_7.msg	1,068	07-26-2022 03:43	c1915b56
pust021_case_7.out	733,917	07-26-2022 04:40	f5f1998e
pust021_case_8.inp	591	11-04-2019 19:43	d0ad7f64
pust021_case_8.msg	1,068	07-26-2022 03:43	49a34f06
pust021_case_8.out	734,318	07-26-2022 04:19	52c2ce7d
pust021_case_9.inp	592	11-04-2019 19:43	52aa51ef
pust021_case_9.msg	1,068	07-26-2022 03:44	adaddd3e
pust021_case_9.out	734,129	07-26-2022 04:37	99ad2033
pust021_case_10.inp	729	11-04-2019 19:43	1875010b
pust021_case_10.msg	1,069	07-26-2022 03:42	57c6ea00
pust021_case_10.out	774,645	07-26-2022 06:05	44e9a272
pust024_case_1.inp	2,059	07-25-2022 12:11	1c16a081
pust024_case_1.msg	2,035	07-26-2022 03:44	f3d54034
pust024_case_1.out	682,454	07-26-2022 04:36	f188d43d
pust024_case_2.inp	2,052	07-25-2022 12:11	540985ef
pust024_case_2.msg	2,036	07-26-2022 03:56	5ba4cee7
pust024_case_2.out	682,328	07-26-2022 04:47	9cd0d747
pust024_case_3.inp	2,056	07-25-2022 12:11	7d85d42c
pust024_case_3.msg	2,036	07-26-2022 04:20	1f9fa86d
pust024_case_3.out	681,851	07-26-2022 05:02	223ce5fd
pust024_case_4.inp	2,056	07-25-2022 12:11	a87bf974
pust024_case_4.msg	2,036	07-26-2022 04:22	cf81c8ff
pust024_case_4.out	681,952	07-26-2022 05:14	b5c4c7be
pust024_case_5.inp	2,053	07-25-2022 12:11	58f623af
pust024_case_5.msg	2,036	07-26-2022 04:22	c56f1303
pust024_case_5.out	681,851	07-26-2022 05:03	054c58a5
pust024_case_6.inp	2,073	07-25-2022 12:11	7ab8bb7a
pust024_case_6.msg	2,036	07-26-2022 04:23	ed2281bf
pust024_case_6.out	681,954	07-26-2022 05:04	01b0ae0e

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Name	Size	Last write date	CRC32
pust024_case_7.inp	2,073	07-25-2022 12:11	1e24c47c
pust024_case_7.msg	2,036	07-26-2022 04:24	2f7812b7
pust024_case_7.out	682,372	07-26-2022 05:07	8e49154e
pust024_case_8.inp	2,073	07-25-2022 12:11	fa933767
pust024_case_8.msg	2,036	07-26-2022 04:25	3e7659c6
pust024_case_8.out	681,688	07-26-2022 05:09	b53ee58b
pust024_case_9.inp	2,069	07-25-2022 12:11	066d5918
pust024_case_9.msg	2,035	07-26-2022 04:37	b6917b9c
pust024_case_9.out	682,620	07-26-2022 05:11	c763600a
pust024_case_10.inp	2,068	07-25-2022 12:11	d623537b
pust024_case_10.msg	2,038	07-26-2022 03:45	970c7d24
pust024_case_10.out	681,856	07-26-2022 04:38	e2cc0901
pust024_case_11.inp	2,070	07-25-2022 12:11	dff955c4
pust024_case_11.msg	2,038	07-26-2022 03:47	dea7f22a
pust024_case_11.out	680,897	07-26-2022 04:43	1b4b8c9a
pust024_case_12.inp	2,070	07-25-2022 12:11	c845af4d
pust024_case_12.msg	2,038	07-26-2022 03:47	293c3fdb
pust024_case_12.out	681,743	07-26-2022 04:42	66354e12
pust024_case_13.inp	2,070	07-25-2022 12:11	bb013fc8
pust024_case_13.msg	2,038	07-26-2022 03:48	70b27ce2
pust024_case_13.out	681,559	07-26-2022 04:43	cf54cd8c
pust024_case_14.inp	2,077	07-25-2022 12:11	1cc0e150
pust024_case_14.msg	2,038	07-26-2022 03:49	a1e8c426
pust024_case_14.out	681,927	07-26-2022 04:41	dcbe453b
pust024_case_15.inp	2,076	07-25-2022 12:11	dd86fa08
pust024_case_15.msg	2,038	07-26-2022 03:50	1e04439f
pust024_case_15.out	681,316	07-26-2022 04:41	52344cae
pust024_case_16.inp	2,076	07-25-2022 12:11	d208ade6
pust024_case_16.msg	2,038	07-26-2022 03:50	d058721f
pust024_case_16.out	682,212	07-26-2022 04:41	41971635
pust024_case_17.inp	2,073	07-25-2022 12:11	74450282
pust024_case_17.msg	2,038	07-26-2022 03:51	66f70aa0
pust024_case_17.out	682,063	07-26-2022 04:43	21c91022
pust024_case_18.inp	2,076	07-25-2022 12:11	ccdd4c5b
pust024_case_18.msg	2,038	07-26-2022 03:53	cbfc62ec
pust024_case_18.out	681,608	07-26-2022 04:52	9cf6b4c1
pust024_case_19.inp	2,076	07-25-2022 12:11	6452e4e2
pust024_case_19.msg	2,038	07-26-2022 03:55	81ef6901
pust024_case_19.out	682,309	07-26-2022 04:55	e7a00663
pust024_case_20.inp	2,072	07-25-2022 12:11	1368bfff
pust024_case_20.msg	2,038	07-26-2022 03:56	19bc5af2
pust024_case_20.out	681,534	07-26-2022 05:00	8af0f7c4
pust024_case_21.inp	2,071	07-25-2022 12:11	c0c70cc0
pust024_case_21.msg	2,038	07-26-2022 04:02	ec6f0e95
pust024_case_21.out	681,725	07-26-2022 05:06	57e1a463
pust024_case_22.inp	2,072	07-25-2022 12:11	f28a48ee
pust024_case_22.msg	2,038	07-26-2022 04:15	6bd1621d
pust024_case_22.out	682,110	07-26-2022 05:19	da0a7d36
pust024_case_23.inp	2,072	07-25-2022 12:11	b7b082d4
pust024_case_23.msg	2,038	07-26-2022 04:20	2332dced
pust024_case_23.out	681,853	07-26-2022 05:12	5e9e982d